Musings on Markets

My not-so-profound thoughts about valuation, corporate finance and the news of the day!

Wednesday, July 29, 2026

Information Timing and Release: The Gaming of Guidance!

I am a lapsed academic, insofar as I have not submitted a paper for publication in more than two decades, but to acquire academic status, I did have to earn a PhD in the distant past. My doctoral dissertation, which like most doctoral theses is little noted and long forgotten, was completed in 1984 and focused on how the frequency of and delays in the release of information plays out in stock price volatility, skew and jumps. I don't plan to rehash that paper, but there have been two long running news stories that reminded me of it. The first is a proposal being floated by the Securities Exchange Commission (SEC) to replace quarterly reporting of financial statements by companies with semi-annual reporting. The second is the opinion voiced by Kevin Warsh, the new Fed Chair, that the Fed should provide less guidance to financial markets on future decisions.

The two stories may seem unconnected, but they have two common features. First, both actions (removing quarterly reporting requirements and reducing/eliminating Fed guidance), if carried through, will remove "news" that markets have become used to receiving and using to calibrate prices. Second, the arguments for and against each of these proposals have parallels. The advocates for removing quarterly reports argue that they *feed into market myopia and increase short-termism in markets*, and the supporters of less Fed guidance believe that this *guidance creates gaming among traders and investors, increasing focus on the FOMC actions at the expense of fundamentals*. The pushback against both proposals comes from those who believe that withholding quarterly reports and Fed guidance *removes information that markets use to set prices, making these prices more volatile and less informative*. As is almost always the case with these debates, there is both truth and hyperbole on both sides, and I will try to thread the needle.

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Earnings Reports

Most investors and traders in equity markets, and especially so in the United States, have spent their investing lifetimes in an environment where companies not only release full financial statements every quarter, but do so with fan fare. As we will note in this section, that has not always been the case, even in the US, and has more recent origins, with setbacks, in many foreign markets.

The History of Earnings Reporting Periodicity

The Securities Exchange Act of 1934, which created the SEC, set up foundational annual reporting requirements (10-Ks) for publicly traded companies, modified in 1955 to require semi-annual reporting and in 1970, quarterly reports, within 45 days from the end of each quarter. That said, there have been forces that have induced firms to report earnings on a more frequent basis to investors well before these regulatory requirements were put in place. The first were the *stock exchanges* that imposed their own constraints, with the *NYSE requiring most firms to report on a quarterly basis as early as 1939*. The second was the recognition by firms that financial transparency (in the form of more frequent and more detailed financial reports) could make them more attractive to investors. As a consequence, it is estimated that in 1931, prior to either the SEC or NYSE mandating disclosure, more than 60% of publicly traded companies were already disclosing information on a quarterly basis.

The shift to more frequent reporting was slower in the rest of the world and has seen more reversals. Europe, for much of the last century, has a patchwork of rules, with some countries adopting stricter disclosure laws than others. The UK imposed mandatory quarterly earnings reporting in 2007, but allowed a shift back to semi-annual reporting in 2014, and the EU also followed a similar timeline, introducing quarterly reports in 2007 and withdrawing that requirement in 2013. In 2003, Singapore started requiring quarterly reporting for firms with market capitalization exceeding S$20 million, but in 2020, shifted away to requiring it only for a subset of firms with financial and regulatory concerns. Japan started its quarterly reporting requirements in 2003 as well, but it too reversed that requirement in 2024. In emerging markets, there are large variations across countries. In India, publicly traded companies are required to report their financials on a quarterly basis, and the same is true for many Brazilian and Chinese companies. In Africa, Nigeria requires quarterly reporting but South Africa has a semi-annual reporting mandate, though many companies voluntarily release quarterly financials; much of the rest of Africa has semi-annual reporting requirements.

In sum, the belief at the start of the twenty first century that the rest of the world would follow the US model of mandated quarterly reporting for publicly traded companies has not come to fruition, as many parts of the world have experimented with mandatory quarterly reporting, before abandoning it in favor of semi-annual reporting, for a variety of reasons. That said, it is worth noting that a significant percentage of firms voluntarily report their financial results on a quarterly basis, even when not mandated, albeit with different degrees of depth.

About Me



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I am a Professor of Finance at the Stern School of Business at NYU. I teach classes in corporate finance and valuation, primarily to MBAs, but generally to anyone who will listen.

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My web site

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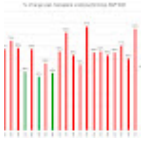
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The Content of Earnings Reports

The debate about how frequently companies should report their financials misses a key detail related to what their financial reports include as content. Focusing on the US, for instance, the magnitude of quarterly earnings reports has increased over time, expanding from bare bones financial statements fin the 1970s to much larger documents that go well beyond financial statements today. In 1980, for instance, a typical quarterly earnings report contained 2000-5000 words, but by the turn of this century, those reports had tripled or quadrupled in size, and the trends continue. The graph below, for instance, looks at the growth in word count for the median quarterly and annual reports in the Russell 3000 companies between 2006 and 2020 (for quarterly) and 1994 to 2020 (for annual):



As you can see the number of words in both quarterly and annual reports has increased over time, and the bulking up of earnings reports can be explained by multiple factors:

1. Accounting rule changes: Accounting rule writers have been busy adding more items to the list of required disclosures for public companies in the last few decades. Some of this increased disclosure (stock-based compensation, for example) reflects a changing business world and is merited, some is in reaction to a corporate scandal and if often knee-jerk and some, in my cynical vie, reflects accounting trying to be relevant to markets again.
2. Macro events: In years of market crises, economic or political, you will see disclosures increase. In the graphs above, notice the spikes in 2008/2009 and 2020, the first in response to the 2008 banking crisis and the latter to COVID. Superimposing the effects of globalization, where a company finds itself exposed to problems in every corner of the world, it has added to the disclosure bloat.
3. Legal Protection: One of the culprits responsible for disclosures bulk is the risk exposure section, where companies are required lay out an exhaustive (and exhausting) list of things that can go wrong in their business models. I have never found a risk disclosure useful in a valuation, as it seems to be written by lawyers with the objective of providing legal cover.
4. Guidance: In the 1980s, quarterly earnings reports were focused on reporting on operations during the quarter in question and management was not expected to, and did not provide, guidance about future quarters. That started to change in the 1990s,



science, and I argue that it is neither; it does not hav...



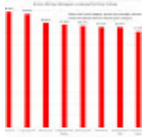
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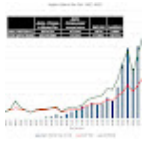
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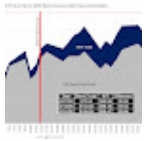
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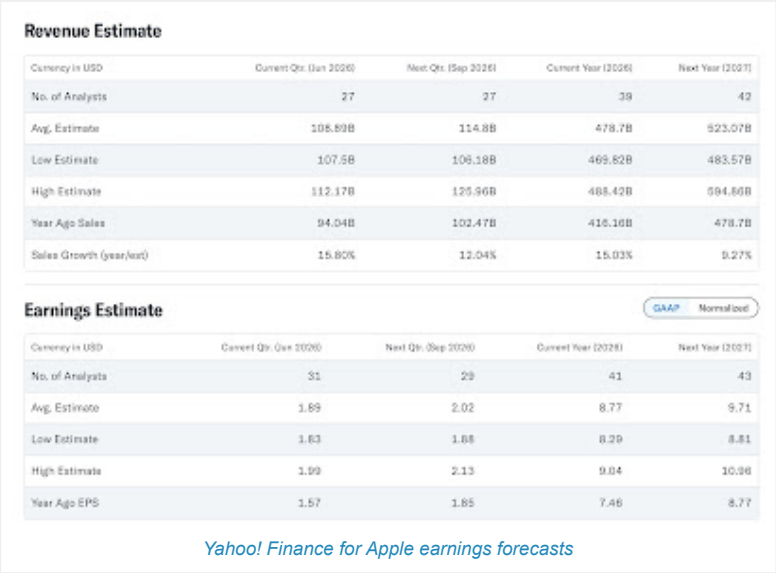
Markets play an expectations game, and in March 2026, we saw the process play out, with all of its upsides and downsides. The month star...

especially with the passage of the 1995 Safe Harbor Law and Reg FD (which prevented companies from selectively leaking information to analysts), and surged through the second half of the decade, peaking in 2003, when more than 50% of all companies providing earnings guidance. Thankfully, the process has receded, with only a fifth of all firms now providing guidance with earnings reports, but it is undeniable that there is much more forward-looking components to earnings reports than used to be the case.

The bulking up of earnings reports is part of a phenomena that I term "disclosure diarrhea" and argue has undercut the usefulness of these reports, with more disclosure perversely making for less information.

The Earnings Game

To make sense of the arguments for and against quarterly earnings reporting, you have to get a measure of what happens leading into and out of these reports, in the "earnings game". The process starts with analysts and investors making forecasts of what the earnings report will contain, almost always including estimates of the earnings per share, but often also containing estimates of expected revenues and even operating metrics (like margins) for high profile companies. The analyst forecasts, at least from sell side analysts, become quasi public information and are often aggregated and reported as **consensus estimates** by financial news services. Zacks, for instance, is one of the services that has been doing this for decades, but that information is now widely accessible on Google and Yahoo! Finance (with the estimates for Apple on July 27, 2026, for the September 2026 earnings report, shown below):



Revenue Estimate

Currency in USD	Current Qtr. (Jun 2026)	Next Qtr. (Sep 2026)	Current Year (2026)	Next Year (2027)
No. of Analysts	27	27	38	42
Avg. Estimate	108.89B	114.8B	478.7B	523.07B
Low Estimate	107.5B	106.18B	469.82B	483.57B
High Estimate	112.17B	125.96B	488.42B	594.86B
Year Ago Sales	94.04B	102.47B	416.16B	478.7B
Sales Growth (year/est)	15.80%	12.04%	15.03%	9.27%

Earnings Estimate

GAAP | Normalized

Currency in USD	Current Qtr. (Jun 2026)	Next Qtr. (Sep 2026)	Current Year (2026)	Next Year (2027)
No. of Analysts	31	29	41	43
Avg. Estimate	1.89	2.02	8.77	9.71
Low Estimate	1.83	1.88	8.29	8.81
High Estimate	1.99	2.13	9.04	10.96
Year Ago EPS	1.57	1.85	7.46	8.77

Yahoo! Finance for Apple earnings forecasts



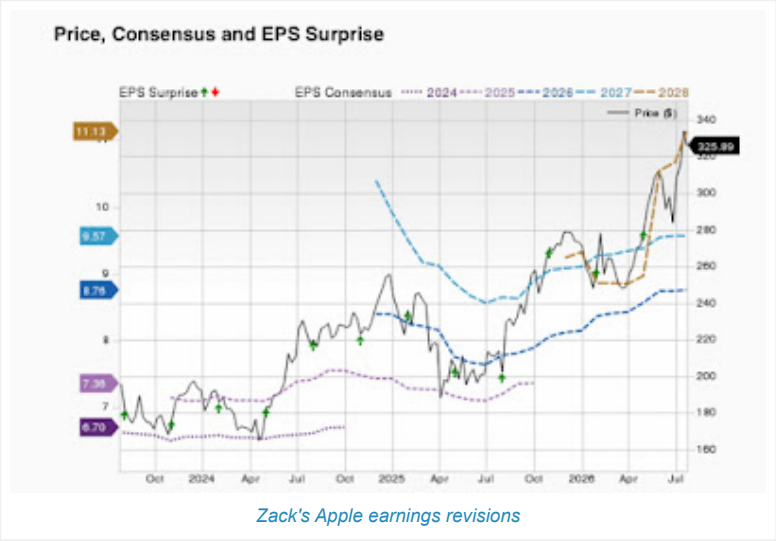
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In the data update posts this year, I have wended my way from the macro (equities collectives, the bond market and other asset classes) ...

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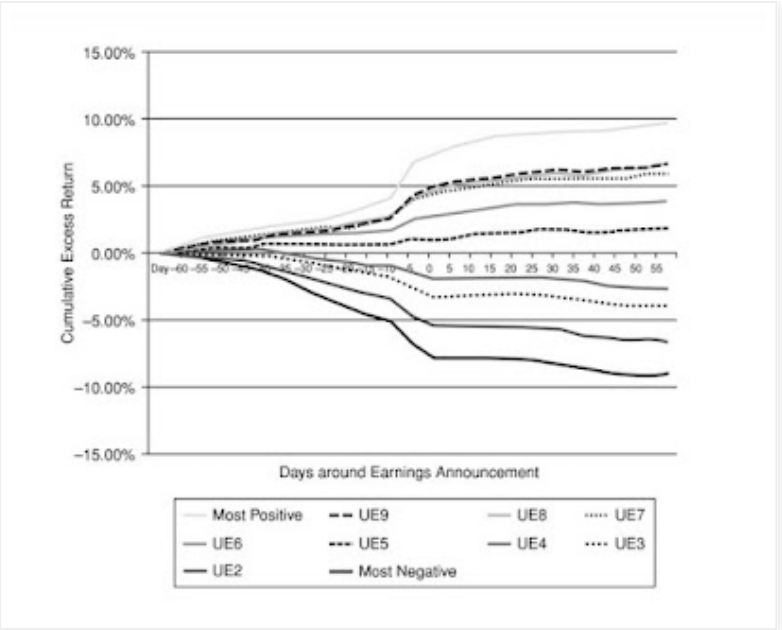
These analyst forecasts, once made, are revisited, partly in response to company-specific news stories and partly to macroeconomic developments, and **revised forecasts** are provided, with services again tracking these revisions for trends (as you can see below for Apple, from Zack's):



Zack's Apple earnings revisions

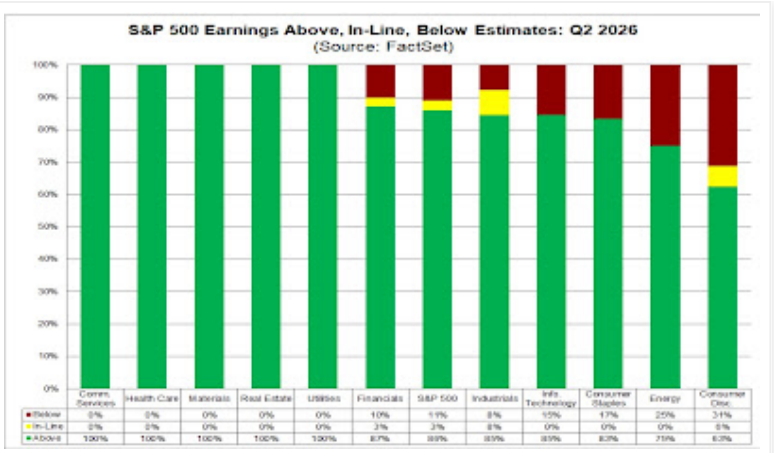
As the earnings release date approaches, analysts continue to revise their estimates, and on the date of the announcement, the actual earnings per share is compared to the expected number, with higher (lower) than expected earnings labeled as positive (negative) surprises. The market price response is often consistent, with positive (negative) earnings surprises translating into increases (decreases) in stock price. The graph below, while dated, looks at stock price responses to earnings surprised classified into ten deciles (from most positive to most negative):

► 2013 (36)
► 2012 (49)
► 2011 (55)
► 2010 (45)
► 2009 (60)
► 2008 (42)



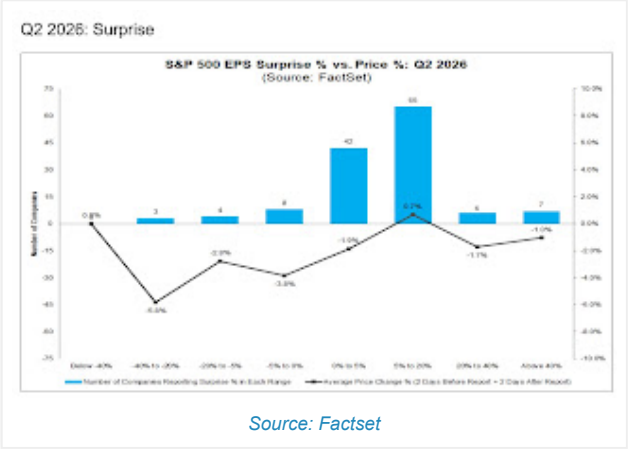
There is some evidence that the market responses to earnings reports have become more muted over time, perhaps because of public access to analyst forecasts and revisions. The link between earnings surprises and stock price changes has become the reason why analysts spend as much time as they do, forecasting earnings per share in the next quarterly report, and why traders focusing on the same metric. There is another aspect of the market reaction to earnings surprises that becomes grist for the trading mill, and it comes from the price drifts in the days after earnings are released, with positive (negative) surprises followed by upward (downward) drifts. While the price drift is small, it may still be large enough to make a difference in active trading, where winning by inches is still winning.

As with almost everything else that is market-related, there are no easy wins in this game, and as more and more people play the earnings forecasting game, new wrinkles have emerged. First, companies have learned to use the flexibility embedded in accounting rules to find ways to beat analyst estimates, with tech companies, in particular, standing out. That earnings gaming plays out as a disproportionately large number of positive earnings surprises (at least among the S&P 500 companies, broken down by sector), as is clear from earnings surprises at the S&P 500 companies in the second quarter of 2026:



Source: Factset

Second, as companies routinely beat analyst estimates, markets readjust, creating the phenomenon of *whispered earnings*, where investors build in the expectation that a company that has historically delivered earnings that are 5% or 10% above estimates will continue to do so, and a lesser number is a negative surprise. In the graph below, I look at the market price reaction to earnings surprises in the second quarter of 2026:



Source: Factset

As you can see, the linkage between earnings surprises and price reaction is weak, with a significant subset of positive surprises resulting in price drops.

The Bottom Line

Much of the debate about whether the US should shift away from quarterly to semi-annual reports can be boiled down to what you think about the time and energy investors and companies spend playing the earnings game, and where that time and energy will be spent in the absence of

quarterly reports. Those who are advocates for less frequent reporting are of the view that the earnings game, focused as it is on next quarter's earnings estimates and whether the company can beat them, contributes to short-termism and distracts from fundamentals. Those who are pushing for preserving the status quo (of quarterly reporting) believe that removing quarterly reports will just shift the game, perhaps more intensively, into the semi-annual reports and that there is value to long term investors from having quarterly reports, gaming notwithstanding.

There is another issue that comes up in the context of quarterly reporting, and what would happen if these reports did not exist. Legal strictures notwithstanding, insiders (from within and outside the firm) trade and make money on material information that they have access to, but the public does not. Removing quarterly reporting will create more of an opening for insiders to make money at the expense of public market investors, and while inside trading may contribute to making prices more informative, it also adds to the sense that financial markets are an unfair game.

I am an investor, and I think that there is a compromise solution that draws on both sides of this argument. I like quarterly reporting for two reasons.

1. There is information in those reports that allows me to update my company valuations, though for many companies, the marginal impact of a quarterly report on value is small.
2. While I have no interest in playing the earnings game, the price corrections that happen around earnings reports serve two purposes. *For companies that I have a position in*, they can operate as catalysts, bringing down (up) the stock price of over valued (under valued) companies. At the same time, almost all of the information that I find useful in an earnings reports is in the financial statements and footnotes, not in the lengthy discussions of risk exposure or in the management guidance, and I would welcome an elimination of these sections and a slimming down of these reports.

Note that none of my arguments for preserving quarterly reporting are about short-termism, and that is intentional. First, I am not sure what short-termism even means, since the cynical answer seems to be that any market movement away from your preferred price direction is short term, and any movement in your favor is indicative of market wisdom. Second, I believe that most market participants, and this is true across time and markets, trade to make money in the near term, and that there is nothing that regulators or rule writers can do to alter this dynamic. In fact, the magic of markets is that millions of trades motivated by opportunism and the short term can still yield a price that is long term and rational. Finally, it remains true that if we were all long-term investors who traded only when the fundamentals drove us to do so, markets would be less liquid and transactions costs would increase; short term traders provide a market service and supply liquidity that we all (including long term investors) benefit from.

I hope that the SEC preserves the current quarterly reporting requirement, while scaling back the volume of disclosure, but if it decides otherwise, it will not materially change much of what I do. I will miss the quarterly updates more with younger, higher-growth firms, where the operating

metrics (revenues, margins etc.) can change quickly over short periods, but it is my guess that many of these firms will voluntarily continue the quarterly reporting tradition.

Fed Guidance on Rates

For most investors who started investing after 2008, the Fed, in particular, and central banks, in general, have loomed large in the investing process. Many investors attribute the low interest rates after 2008 almost entirely to Fed actions, and by extension, blame the Fed for the higher rates since 2022. I have long argued that not only is this perception incorrect, but that it is unhealthy for investors to view the Fed as either savior or villain.

A Short (and Personal) History of the Fed

I started in equity markets in the 1980s, when Paul Volcker as the chair of the Fed played a central role in getting inflation back into check. I might have been ignorant, but I did not know the names of any of the members of the Federal Open Market Committee and had no idea when they met. Changes in the fed funds rate, the only rate effectively controlled by the FOMC, would percolate their way into markets, but I don't remember them being central to equity market movements.

Volcker was followed by Alan Greenspan, and while he acquired rockstar status (at least among investors) in the late 1990s, his views on rates were superseded by his views on equity investors (and their irrational exuberance). The FOMC met eight times per year during that period, and you can access the meeting minutes and actions on the Fed website here, but it stayed away from explicit guidance about future rate changes, choosing to send subtle hints instead.

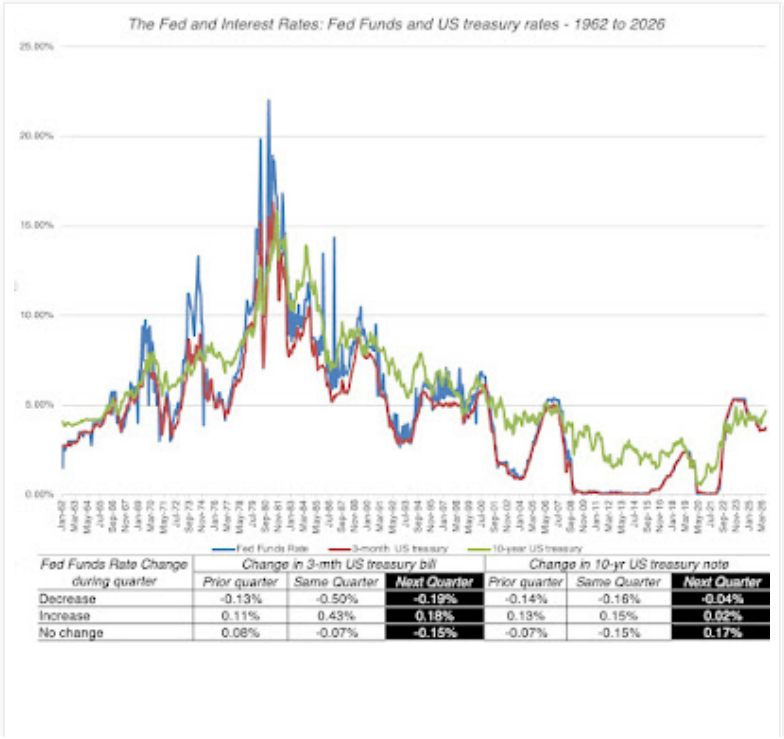
The sea change in Fed behavior and centrality occurred with the 2008 market crisis, when the Fed first introduced explicit guidance noting that rates would stay low "for some time", and it has largely continued that practice through the stewardships of Bernanke (2006-2014), Yellen (2014-2020) and Powell (2020-2026). Along the way, its place in markets has changed, as both bond and equity investors have become focused on the Fed as the arbiter of interest rates and director of the economy.

The Fed's Powers (and Powerlessness)

To understand the extent and limits of the Fed's capacity to guide rates and the economy, it is useful to begin with an understanding of what it does. Through its twelve districts that span the United States, the Fed collects information on almost every aspect of the economy, from price pressures building on consumers and producers to the pace of economic growth. While there are other government agencies that also track these statistics, it is undeniable that the Fed has a big picture view and access to more data than any other government agency. The Federal Open Market Committee, composed of all of the members of the board of governors and representatives of the district presidents, sets Fed policy on open market operations (where the

Fed buys and sells US government securities), the size of the Fed's balance sheet and the Fed Funds rate (an overnight rate at which banks can borrow and lend their reserves). In addition to the FOMC providing policy direction on inflation and the economy, the Fed chair testifies to Congress every six months, facing and answering questions from legislators.

As the key interest rate set by the Fed, *the Fed Funds rate* often acquires an outsized role and there are good reasons to pay attention to it. First, it *operates as a signal* of what the Fed is seeing in the data it has collected on the economy, with an increase (decrease) in rates indicating that it sees higher (lower) inflation and an overheated (slowing) economy. Second, *there are interest rates that are directly tied to the Fed Funds rate*, where changes percolate down to businesses and customers; the prime rate and some credit card and CD rates move with the Fed Funds rate. That said, I believe that Fed's capacity to affect interest rates is far more limited than most believe, for two reasons. First, while there is positive correlation between Fed Funds rates and short-term market-set rates (like the US treasury bill rate), there is as much evidence (if not more) that the latter lead the former, rather than the other way around. Put simply, Fed funds rates tend to be increased (cut) after short term treasury rates have gone up (down), suggesting that the Fed is mimicking the market. Second, the relationship between Fed Funds rates and long-term market-set rates, which drive asset valuation and affect borrowers more, is even weaker. To back these contentions, I chart the effective fund funds rate, the three-month US treasury bill rate and the 10-year treasury note rate on a monthly basis from January 1962 to June 2026:



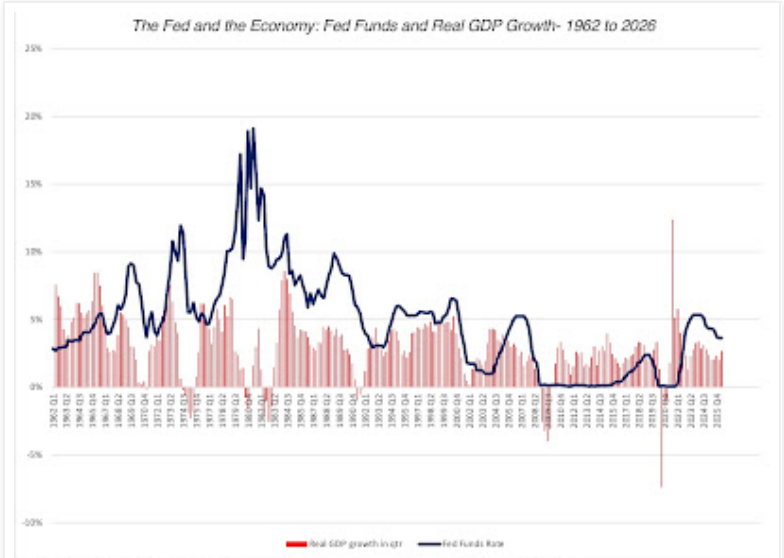
Fed Funds Rate Change during quarter	Change in 3-mth US treasury bill			Change in 10-yr US treasury note		
	Prior quarter	Same Quarter	Next Quarter	Prior quarter	Same Quarter	Next Quarter
Decrease	-0.13%	-0.50%	**-0.19%**	-0.14%	-0.16%	**-0.04%**
Increase	0.11%	0.43%	**0.18%**	0.13%	0.15%	**0.02%**
No change	0.08%	-0.07%	**-0.15%**	-0.07%	-0.15%	**0.17%**

Fed Funds Rate Change during quarter	Real GDP growth rate		
	Prior quarter	Same Quarter	Next Quarter
Decrease	2.42%	2.36%	2.24%
Increase	3.50%	3.51%	3.61%
No change	3.26%	4.05%	3.38%

Download data

At the bottom of the graph, I have a table where I look at the data on a quarterly basis, and break it down into three groups - quarters where the fed funds rate decreased, quarters where it increased and quarters where it stayed unchanged. With both fed funds rate increases and decreases, *you can see that the link with short term rates is stronger*, and with both short term and long term rates, *the bulk of the change in rates happens prior to or in the quarter that the Fed Funds rate changed*, but there is only a mild spill over into the quarter after, with three month rates, and almost no spillover, with long term rates. Put simply, baed on this history, it looks like changes in fed funds rate are less signals of future movements in interest rates and more reflectors of changes that have already happened.

The Fed's weaknesses in setting interest rates also plays out in its capacity to alter the trajectory of the real economy. While there are clearly periods that you can point to where Fed actions have had a material impact on he economy, with the Fed Fund rate was hiked to 20% under Paul Volcker in 1981, and triggering a deep recession, being a prime example, the link between Fed Fund rates and economic growth remains tenuous. In the graph below, I look at the changes in Fed Funds rates and real GDP growth in the quarter leading into, the quarter of and the quarter after the change:

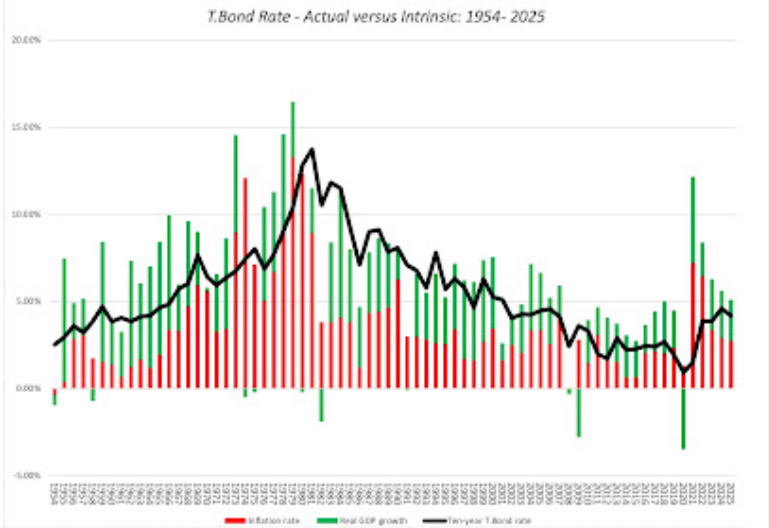


Fed Funds Rate Change during quarter	Real GDP growth rate		
	Prior quarter	Same Quarter	Next Quarter
Decrease	2.42%	2.36%	2.24%
Increase	3.50%	3.51%	3.61%
No change	3.26%	4.05%	3.38%

Download data (FRED)

Again, there is little backing for conventional wisdom, which is that fed tightening (by raising the Fed funds rate) leads to drops in real growth (or even recessions) and that fed loosening (by lowering the Fed funds rate) is a signal of higher economic growth in the future. In fact, the more general conclusion that one can draw from the data is that the fed effect on the real economy has been more "meh" than "wow".

The gap between investor perception on what the Fed can control on interest rates and the economy and its actual powers is not just wide, but potentially dangerous. From a policy perspective, it can lead to perverse actions, where central banks are pressured to lower the rates they control (like the Fed Funds rate) in the face of high inflation, leading to even higher inflation in the future. From an investor and business perspective, the focus on what the Fed is doing or will do can take attention away from the fundamentals, especially inflation, that ultimately drive both interest rates and growth.



Year end	Ten-year T.Bond rate	Inflation rate	Real GDP growth	Intrinsic riskfree rate	Intrinsic - T.Bond Rate
1954-1980	5.83%	4.49%	3.50%	7.98%	-2.15%
1981-2008	6.88%	3.26%	3.04%	6.30%	0.58%
2009-2020	2.36%	1.81%	1.32%	3.13%	-0.77%
2021-2025	3.61%	4.52%	2.98%	7.50%	-3.90%
1954-2025	5.50%	3.57%	2.92%	6.48%	-0.98%
End of 2025	4.18%	2.74%	2.36%	5.10%	-0.92%

Download data

As you can see, much of the variation in long term interest rates (with the ten-year US treasury rate standing in as proxy) can be explained by movements in inflation and real economic growth

over time, not Fed action or inaction.

The Warsh Doctrine?

All Fed chairs have had to wrestle with the problem of being perceived as all-powerful, when their true powers are limited, but Kevin Warsh is perhaps more exposed than any of his predecessors. The market fixation with the Fed is now deeply embedded in market, and there are politicians on both sides of the aisle who seem to think that Warsh can bring rates (mortgage, treasury) down to 2% or lower, if he so desires, when the truth is that with inflation expectations running at 2.5-3%, there is no chance of that happening.

The pathway out of this problem will be long and there will be pushback, but the end game should be a world where you see and hear from the Fed less, not more. I do believe that the decision to reduce or withhold guidance is a good first step, and it has to be followed by more open humility from the Fed (and from Warsh) about the limits of its powers and honesty about how frequently it follows markets, rather than leads them. In the context of today's (July 29, 2026) decision by the FOMC to leave rates unchanged, for instance, the subtext is that while inflation is running hotter than desired (3% or more, as opposed to the targeted number of 2%), much of that inflation is being driven by a war and its effect on oil prices

There will be some who feel that markets will be lost without Fed guidance, but I don't think so. After all, financial markets set interest rates and stock prices before the guidance era, and did a pretty good job. In fact, I think that the surge in guidance from the Fed has led many in markets to abdicate their responsibility for paying heed to fundamentals and gauging what interest rates should be.

Conclusion

If there is a takeaway from this post, it should be that there is nothing inherently good about having more disclosure. In fact, there is a tipping point, where information overload can cause investors to behave in perverse ways. Thus, I am less of an absolutist about the quarterly versus semi-annual reporting debate than some, though my view is that rather than reduce the frequency of reporting, the SEC should be looking at slimming down reports, by replacing one-size-fits-all disclosure requirements with targeted disclosures and keeping the focus on reporting what has happened rather than prognosticate about the future.

With the Federal Reserve too, I think less is more is a better strategy - less guidance from the Fed about what it will do in the future, less opining from FOMC members about interest rates and the economy and less attention to FOMC meetings and the smoke signals that emerge from these meetings. Markets will step in to fill the vacuum, and that is good not just for investors but for the Fed, since its decisions are informed by those market judgments.

YouTube Video



Information Timing and Release: The Debate over

Aswath Damodaran





Watch on

Data

1. Fed Funds Rates and US Treasury rates, by month (1962-2026)
2. Fed Funds Rates and Real GDP growth, by quarter (1962-2026)
3. Intrinsic risk free rates and 10-year Treasuries (1954-2025)

Posted by Aswath Damodaran at 6:06 PM No comments:

Labels: Earnings Reports, Information, The Fed

Wednesday, July 15, 2026

Country Risk: Drivers, Measures and Investment Implications - The 2026 Edition!

I am a creature of habit in my personal and professional life, and in the context of the content that I post online, there is a ritual that I follow with my data updates. I start the year with my general data update online, and follow up with a series of posts where I examine the implications of this data for investing and corporate finance. Since 2008, I have also done annual update papers on equity risk premiums in March of each year, with the link to the 2026 update here, and

country risk in July of each year, where I look at the topics in more details, trying as best as I can to integrate the data, research and my own thinking. This year's country risk update paper is now available, and as in prior years, I will spend this post looking what causes risk to vary across countries, how to measure those risk variations and the implications for businesses and investors.

Country Risk: Relevance

In my years as a business school student, country risk was given short shrift and I don't remember spending much time talking or thinking about it. Part of the reason was that business school education was dollar-centric and built on the presumption that most graduates would go to work in New York, London or Tokyo, and have little need to confront country risk on a day-to-day basis. For those who raised country risk as an issue, the response was that you could, as a company or investor with global exposure, diversify it away. Both presumptions were wrong even then, and have become even more flawed over time as we have sold both companies and investors on the benefits of globalization.

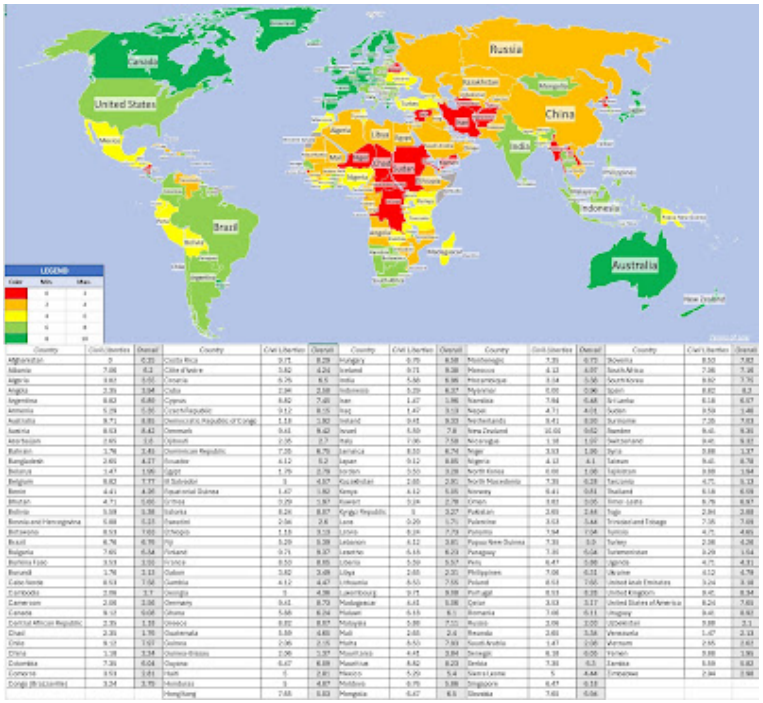
For businesses, the exposure to country risk comes from both the revenue side, as larger portions of every company's revenues come from foreign markets, and the cost side, as production gets outsourced to locales overseas. That exposure tends to increase as companies scale up, and is higher in some sectors than others; technology companies, for instance, get far more of their revenues from other non-domestic markets than manufacturing or service businesses. Outside of utilities (power, water), it is rare for a company to be entirely domestic-focused on both its revenue and cost sides. For investors, the initial draw of investing in foreign markets might have been diversification but the greater pull has come from greed, i.e., the belief that you can higher returns in the rest of the world. That process was accelerated by the creation of investment vehicles (index and mutual funds) that made investing overseas easier, the lowering of transactions costs across markets and a greater standardization of financial statements and disclosure across the globe. The home bias in portfolios, i.e., the skewing of portfolios towards domestic market investments, has not disappeared but it is lower than it was at the turn of the last century.

The notion that country risk is diversifiable, i.e., that if you are operating or investing across the world, the risks will average out across countries, has been undercut by the increased correlation across global equity markets, and especially so during market crises (which is when you care the most). At the risk of being hyperbolic, there is no place to hide from country risk, for either businesses or investors, and ignoring or dismissing country risk is not an option. I discovered this truth in the 1990s, when I found myself in need of a mechanism to incorporate country risk into my corporate financial analysis and valuations, and the process that you see described in this post was born from that need. I would hasten to add that the process that I describe has very little intellectual firepower behind it, puts pragmatism ahead of theory and most importantly is a work-in-process.

Country Risk: Drivers

I don't think that there would be much disagreement, if I assert that it is riskier to invest in some parts of the world than others, but there is likely to be plenty of disagreement on why there are risk differences and which parts of the world are riskiest. In the broadest sense, I argue that variation in business risk across countries can be traced to four factors - the *political structure* of the country (democracy vs authoritarian), the *prevalence of corruption* in the country (operating as a hidden tax and distorting business outcomes), the *extent of violence* in the country (from internal and external forces) and the *strength of the legal system* in enforcing property rights and contractual obligations.

On the political risk front, I looked at the **EIU's Democracy Index,** a composite score measuring both political freedom and protections of civil liberties, with the caveat that any index that tries to measure these will make subjective judgements that not everyone will agree with. In their most recent update, here is what the EIU scores looked like around the world:



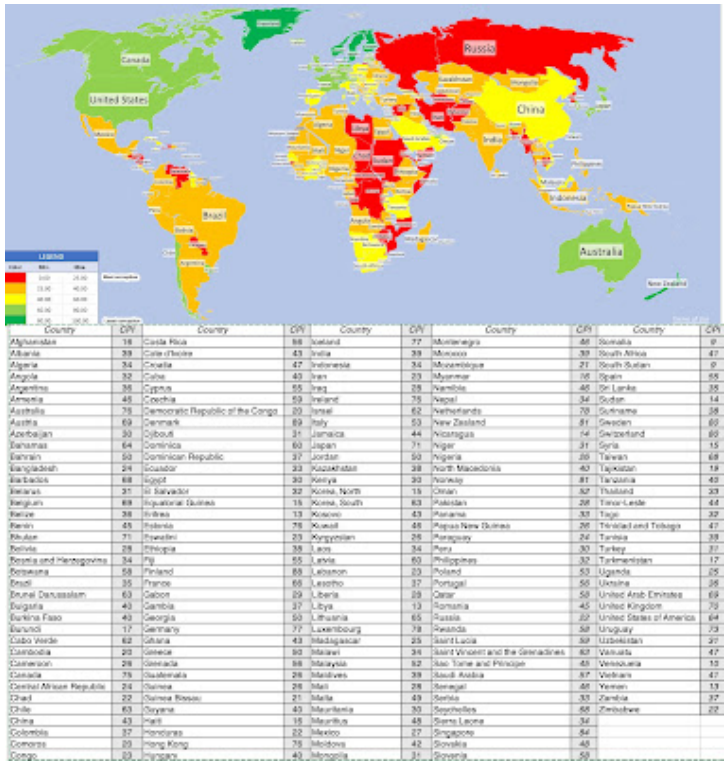
Source: Economist

Low (High) score: Least (Most) freedom

Based on these scores, the tilt towards authoritarianism has increased over the last decade, with only 7.3% of the world's population living in democracies at the end of 2025. Note, though, that there is still an open question of whether businesses and economies do better under democratic than authoritarian regimes, and the answer in the research is at best a "maybe". From a risk

perspective, democratic regimes create more continuous risk for businesses, with elections bringing regulatory and rule changes to economies, than authoritarian regimes, where governments can promise more continuity in policy, but when change does come to the latter, it is more likely to be large and wrenching.

Corruption is a fact of life in much of the world, and businesses often have no choice but to pay the price to survive and grow. **Transparency International,** a global coalition against corruption, tries to capture the extent of corruption, comes up with corruption scores for countries, with lower scores indicating less corruption, and the most recent edition contains the following:



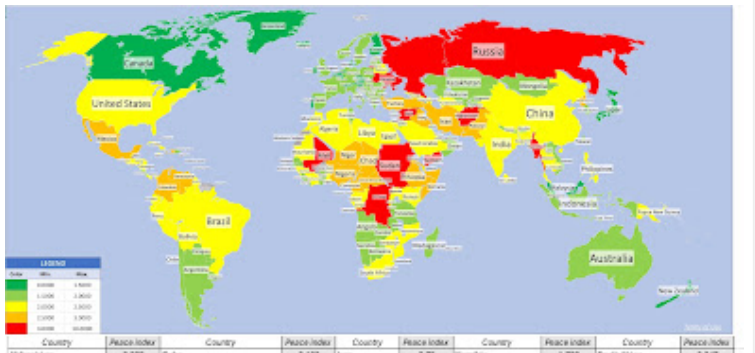
Country	CPI	Country	CPI	Country	CPI	Country	CPI	Country	CPI
Afghanistan	16	Costa Rica	58	Iceland	72	Montenegro	46	Somalia	9
Albania	39	Cote d'Ivoire	43	India	39	Morocco	39	South Africa	47
Algeria	34	Croatia	47	Indonesia	34	Mozambique	27	South Sudan	9
Angola	32	Cuba	46	Iran	23	Myanmar	28	Spain	58
Argentina	38	Cyprus	55	Iraq	20	Namibia	48	Sri Lanka	38
Armenia	45	Czechia	59	Ireland	75	Nepal	34	Sudan	14
Australia	78	Democratic Republic of the Congo	20	Israel	62	Netherlands	79	Suriname	38
Austria	69	Denmark	88	Italy	53	New Zealand	87	Sweden	85
Azerbaijan	30	Djibouti	31	Jamaica	44	Nicaragua	14	Switzerland	85
Bahamas	64	Dominica	60	Japan	73	Niger	31	Syria	13
Bahrain	50	Dominican Republic	37	Jordan	50	Nigeria	26	Taiwan	68
Bangladesh	24	Ecuador	33	Kazakhstan	38	North Macedonia	40	Tajikistan	18
Barbados	68	Egypt	30	Kenya	30	Norway	84	Tanzania	40
Belarus	35	El Salvador	32	Korea, North	15	Oman	62	Thailand	36
Belgium	69	Equatorial Guinea	16	Korea, South	63	Pakistan	28	Timor-Leste	44
Belize	38	Eritrea	13	Kosovo	43	Panama	33	Togo	32
Benin	45	Estonia	75	Kuwait	46	Papua New Guinea	26	Trinidad and Tobago	47
Bhutan	71	Eswatini	23	Kyrgyzstan	26	Paraguay	24	Tunisia	39
Bolivia	28	Ethiopia	38	Laos	34	Peru	30	Turkey	37
Bosnia and Herzegovina	34	Fiji	55	Latvia	60	Philippines	32	Turkmenistan	17
Botswana	58	Finland	85	Lebanon	23	Poland	53	Uganda	25
Brazil	35	France	68	Lesotho	37	Portugal	56	Ukraine	26
Brunei Darussalam	63	Gabon	29	Liberia	29	Qatar	58	United Arab Emirates	69
Bulgaria	43	Gambia	37	Libya	13	Romania	45	United Kingdom	78
Burkina Faso	40	Georgia	58	Lithuania	65	Russia	22	United States of America	64
Burundi	17	Germany	77	Luxembourg	78	Rwanda	58	Uruguay	73
Cabo Verde	62	Ghana	43	Madagascar	25	Saint Lucia	59	Uzbekistan	30
Cambodia	20	Greece	50	Malawi	34	Saint Vincent and the Grenadines	62	Vanuatu	47
Cameroon	26	Grenada	58	Malaysia	62	Sao Tome and Principe	45	Venezuela	13
Canada	76	Guatemala	26	Maldives	38	Saudi Arabia	57	Vietnam	47
Central African Republic	24	Guinea	26	Mali	28	Senegal	45	Yemen	13
Chad	22	Guinea Bissau	21	Malta	49	Serbia	33	Zambia	37
Chile	63	Guyana	40	Mauritania	30	Seychelles	66	Zimbabwe	22
China	43	Haiti	16	Mauritius	48	Sierra Leone	34		
Colombia	37	Honduras	23	Mexico	27	Singapore	84		
Comoros	20	Hong Kong	76	Moldova	42	Slovakia	49		
Congo	20	Hungary	46	Mongolia	31	Slovenia	58		

Source: Transparency International
Low (High) score: Most (Least) corruption

Northern Europe has the lowest corruption scores, followed by Canada, United States and Australia, but large portions of Africa have high exposure to corruption, with Latin America and much of Asia falling in the middle.

Living in the midst of violence takes a toll, and that toll is extracted from businesses that try to operate in its presence. **Vision of Humanity** computes peace scores for countries, measuring

exposure to both violence within the country as well as from wars and terrorism. The most recent peace scores are reported below:



Country	Peace Index	Country	Peace Index	Country	Peace Index	Country	Peace Index	Country	Peace Index
Afghanistan	3.220	Cuba	2.123	Iran	2.75	Namibia	1.789	South Africa	2.347
Albania	1.822	Cyprus	1.803	Iraq	2.882	Nepal	1.987	South Korea	1.736
Algeria	2.042	Czechia	1.435	Ireland	1.26	Netherlands	1.481	South Sudan	3.117
Angola	1.987	Democratic Republic of Congo	3.292	Israel	3.185	New Zealand	1.282	Spain	1.678
Argentina	1.788	Denmark	1.393	Italy	1.982	Nicaragua	2.267	Sri Lanka	2.676
Armenia	1.695	Djibouti	2.276	Jamaica	2.047	Niger	2.759	Sudan	3.323
Australia	1.485	Dominican Republic	1.898	Japan	1.44	Nigeria	2.889	Sweden	1.739
Austria	1.294	Ecuador	2.459	Jordan	1.957	North Korea	2.911	Switzerland	1.294
Azerbaijan	2.067	Egypt	2.157	Kazakhstan	1.575	North Macedonia	1.799	Syria	3.354
Bahrain	2.089	El Salvador	2.126	Kenya	2.382	Norway	1.484	Taiwan	1.73
Bangladesh	2.316	Equatorial Guinea	2.004	Kosovo	1.988	Oman	1.738	Tajikistan	1.996
Belarus	2.287	Eritrea	2.542	Kuwait	1.642	Pakistan	2.787	Tanzania	1.968
Belgium	1.498	Estonia	1.559	Kyrgyz Republic	1.988	Palestine	2.811	Thailand	2.017
Benin	2.211	Eswatini	2.094	Laos	1.783	Panama	2.006	Timor-Leste	1.758
Bhutan	1.536	Ethiopia	2.888	Latvia	1.568	Papua New Guinea	2.23	Togo	2.381
Bolivia	2.035	Finland	1.42	Lebanon	2.674	Paraguay	1.981	Trinidad & Tobago	2.02
Bosnia & Herzegovina	1.886	France	1.967	Lesotho	2.287	Peru	2.073	Tunisia	1.998
Botswana	1.743	Gabon	2.238	Liberia	1.989	Philippines	2.148	Turkiye	2.852
Brazil	2.472	Gambia	1.855	Libya	2.476	Poland	1.713	Turkmenistan	2.016
Bulgaria	1.61	Georgia	2.186	Lithuania	1.588	Portugal	1.375	Uganda	2.257
Burkina Faso	3.056	Germany	1.533	Madagascar	1.895	Qatar	1.593	Ukraine	3.434
Burundi	2.674	Ghana	1.898	Malawi	1.965	Republic of the Congo	2.132	United Arab Emirates	1.817
Cambodia	2.059	Greece	1.764	Malaysia	1.889	Romania	1.721	United Kingdom	1.834
Cameroon	2.685	Guatemala	2.174	Mali	3.081	Russia	3.441	United States	2.443
Canada	1.481	Guinea	2.393	Mauritania	2.284	Rwanda	2.036	Uruguay	1.764
Central African Republic	2.953	Guinea-Bissau	2.113	Mauritius	1.586	Saudi Arabia	2.035	Uzbekistan	1.926
Chad	2.985	Guyana	2.149	Mexico	2.636	Senegal	1.936	Venezuela	2.692
Chile	1.886	Haiti	2.731	Moldova	1.958	Serbia	1.814	Vietnam	1.721
China	2.093	Honduras	2.347	Mongolia	1.719	Sierra Leone	1.887	Yemen	3.392
Colombia	2.686	Hungary	1.5	Montenegro	1.686	Singapore	1.387	Zambia	1.814
Costa Rica	1.643	Iceland	1.696	Morocco	2.012	Slovakia	1.669	Zimbabwe	2.223
Cote d'Ivoire	2.096	India	2.329	Mozambique	2.273	Slovenia	1.409		
Croatia	1.559	Indonesia	1.788	Myanmar	3.045	Somalia	2.983		

Source: Vision of Humanity
Low (High) score: Most (Least) peaceful

Canada, Australia, Japan and much of Europe score high on the peace dimension, and while Latin America and Africa score lower, there are portions of each continent that are more peaceful. The Russia-Ukraine war has created a huge area of violence across Eastern Europe and Russia, and exposure to gun violence creates a drag on the United States.

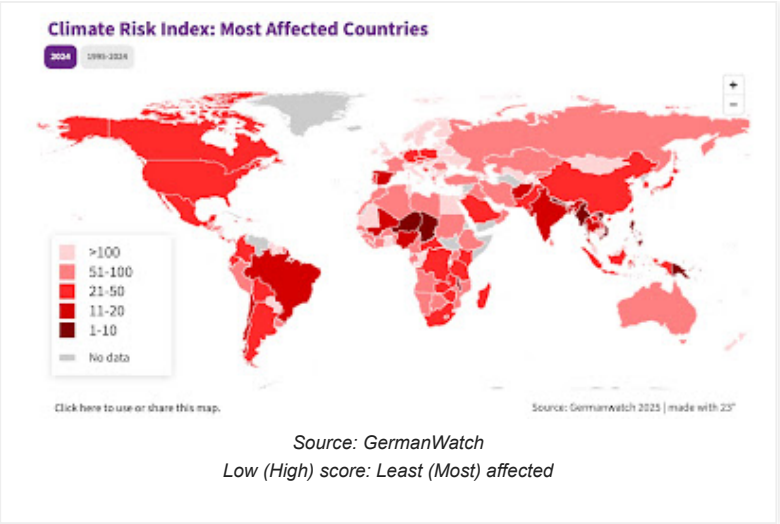
Businesses are dependent on the legal system to enforce property rights as well as contractual obligations. Countries that have legal systems that are either capricious on these fronts, or hopelessly slow in acting, create challenges for businesses that operate in them, creating both costs and risks that they otherwise would not face. Property Rights Alliance is an entity that tracks international property rights across the world, and in their most recent update, their property rights scores by country are captured below:



Source: International Property Rights
Low (High) score: Least (Most) property rights

There are wide differences across regions, when it comes to legal and property rights, with Latin America, Africa and Asia lagging and Europe, Australia and much of North America leading.

There is one final dimension that I have added to country risk in recent years that captures exposure to climate risk. While there are many different entities that measure this exposure, each one with its own skews, the map below which shows the climate risk exposure, by country, from GermanWatch:

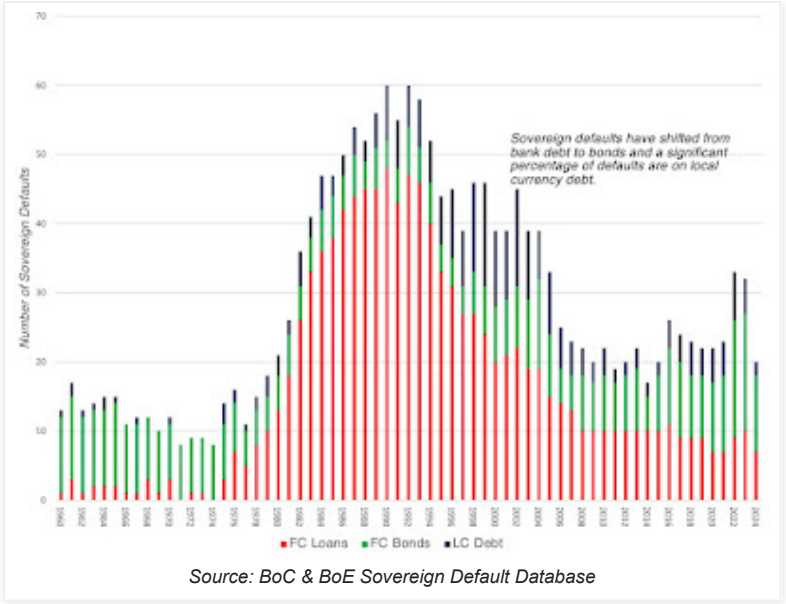


Source: GermanWatch
Low (High) score: Least (Most) affected

There are two reasons why climate risk has not become a bigger topic in country risk discussions. The first is that there is no part of the globe that is unaffected, making it less of a differentiator across countries on the risk dimension. The second is that climate risk, by itself, is an abstraction for businesses, until it starts affecting the bottom line, and while there are individual companies that are being impacted, the aggregate effects, at least at the moment, are not big enough to change the discussion.

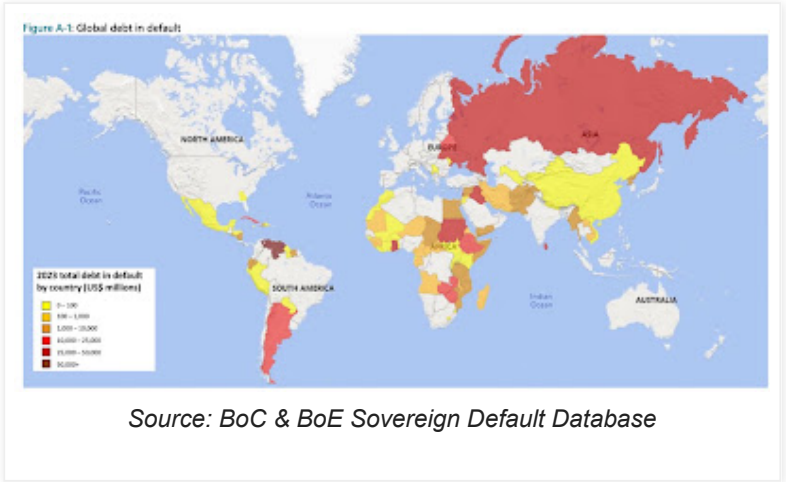
Country Default Risk

While country risk is determined by multiple factors, the challenge that businesses is in consolidating all of those risks into one number. The market that does this most directly is the debt market, where, when countries (sovereigns) seek to borrow money, lenders determine the interest rates to charge them, based upon perceived default risk. To understand why lenders worry about default with sovereign debt, you can start by looking at the history of sovereign defaults in the graph below:



Source: BoC & BoE Sovereign Default Database

Debt defaults, which soared in the 1980s and 1990s, have been lower in this century, with a shift away from loan defaults (where banks are usually the lenders) to defaults in the bond market. It is also worth noting that a *non-trivial portion of sovereign defaults in each year are local currency defaults*, indicating that for some borrowers, the costs of defaulting are viewed as smaller than the costs of inflation arising from printing more currency to pay off debt. Over time, Latin America has been the epicenter for sovereign default, but at the end of 2023, sovereign debt in default had a wide geographical spread:



Source: BoC & BoE Sovereign Default Database

The most widely accessible measures of sovereign default risk remain **sovereign ratings,** with ratings agencies operating as (imperfect) arbiters. At the start of July 2026, the graph below

reports the sovereign ratings for all rated countries, from S&P, Moody's and Fitch:

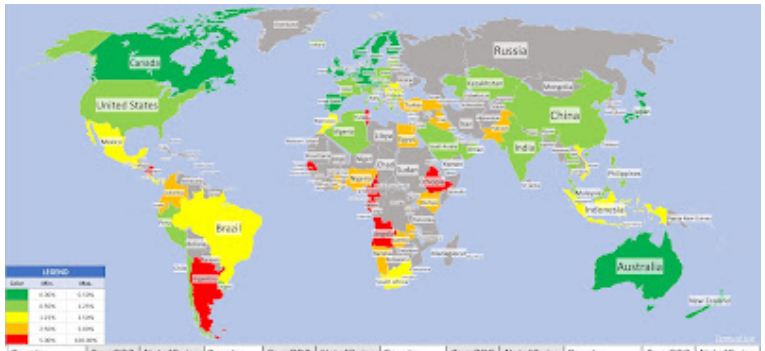


Country	S&P	Moody's	Fitch	Country	S&P	Moody's	Fitch	Country	S&P	Moody's	Fitch	Country	S&P	Moody's	Fitch
Albania	BB	Ba3	NR	Sweden	B-	Caa1	CCC+	Libya	NR	NR	B	Romania	BBB-	Baa3	BBB-
Andorra	A-	Baa1	A-	Egypt	B	Caa1	B	Liechtenstein	AAA	NR	NR	Russia	CC	Ca	RD
Angola	B-	B3	B-	El Salvador	NR	B3	B-	Lithuania	A+	A2	A+	Rwanda	B+	B2	B+
Argentina	B-	Caa1	B-	Estonia	AA-	A1	A+	Luxembourg	AAA	Aaa	AAA	Saint Vincent and the Grenadines		B1	NR
Armenia	BB-	Ba3	BB-	Ethiopia	CCC	Caa2	RD	Macao	NR	Aa3	AA	San Marino	A-	NR	BBB
Aruba	A-	Baa1	BBB+	Fiji	BB-	B1	NR	Madagascar	B-	NR	NR	Saudi Arabia	A+	Aa3	A+
Australia	AAA	Aaa	AAA	Finland	AA+	Aa1	AA	Malawi	NR	NR	B-	Senegal	CCC+	Caa1	NR
Austria	AA+	Aa1	AA	France	A+	Aa2	A+	Malaysia	A-	A3	BBB+	Serbia	BBB-	Ba2	BB+
Azerbaijan	BB+	Ba1	BB+	Gabon	B	Caa2	CCC	Maldives	NR	Caa2	CCC	Seychelles	NR	NR	BB
Bahamas	BB-	Ba1	BB-	Gambia	NR	NR	CCC	Mali	B	Caa2	B-	Singapore	Aaa	Aaa	AAA
Bahrain	B	B2	B	Georgia	BB	Ba2	BB	Malta	A-	A2	A+	Slovakia	A	A3	A-
Bangladesh	BB-	B2	B+	Germany	AAA	Aaa	AAA	Mauritius	BBB-	Baa1	NR	Slovenia	AA	A2	A+
Barbados	B+	C	B-	Ghana	B-	Caa1	B	Mexico	BBB	Baa1	BBB-	Solomon Islands	NR	Caa1	NR
Belarus	CC	C	CCC	Greece	BBB	Baa3	BBB	Moldova	NR	B2	B+	South Africa	BB	Ba2	BB
Belgium	AA-	A1	A+	Grenada	BB-	NR	NR	Mongolia	BB-	B	B+	South Korea	AA	Aa2	AA-
Belize	B-	Caa1	NR	Guatemala	BB+	Ba1	BB+	Montenegro	B+	Ba2	NR	Spain	A+	A3	A
Benin	BB-	B1	B+	Guinea	B+	NR	NR	Mozambique	DDD-	NR	NR	Sri Lanka	CCC+	Caa1	CCC+
Bermuda	A+	A1	NR	Honduras	BB-	B1	B	Morocco	BBB-	Ba1	BB+	Suriname	CCC+	Caa1	C
Bolivia	CCC+	Caa1	CCC+	Hong Kong	AA+	Aa3	AA-	Mozambique	CCC+	Caa2	CC	Swaziland	NR	B3	NR
Bosnia and Herzegovina	B	B3	NR	Hungary	BBB-	Baa2	BBB	Namibia	NR	B1	BB-	Sweden	AAA	Aaa	AAA
Botswana	BBB-	Baa1	NR	Iceland	A+	A1	A+	Nepal	NR	NR	BB-	Switzerland	AAA	Aaa	AAA
Brazil	BB	Ba1	BB	India	BBB	Baa2	BBB-	Netherlands	AAA	Aaa	AAA	Taiwan	AA+	Aa3	AA
Bulgaria	BBB+	Baa1	BBB+	Indonesia	BBB	Baa2	BBB	New Zealand	AA+	Aaa	AA+	Tajikistan	B	B3	NR
Burkina Faso	CCC+	NR	NR	Iran	NR	B2	B+	Nicaragua	B-	B2	B	Tanzania	NR	B1	B+
Cambodia	B	B2	NR	Iraq	B-	Caa1	B-	Niger	NR	Caa2	NR	Thailand	BBB+	Baa1	BBB+
Cameroon	B-	Caa1	B	Ireland	AA+	Aa3	AA	Nigeria	B	B3	B	Togo	B+	B3	NR
Canada	AAA	Aaa	AA+	Israel	A	Baa1	A	North Macedonia	BB-	NR	BB+	Trinidad and Tobago	BBB-	Ba2	NR
Cape Verde	B-	NR	B	Italy	BBB+	Baa2	BBB+	Norway	AAA	Aaa	AAA	Tunisia	NR	Caa1	B-
Chad	NR	NR	B-	Ivory Coast	B1	Ba2	B1	Oman	BB+	Ba1	BBB-	Turkey	B	Ba3	BB-
Chile	A	A2	A-	Jamaica	NR	Ba3	BB-	Pakistan	B-	Caa2	B-	Turkmenistan	NR	NR	BB-
China	A+	A1	A	Japan	A+	A1	A	Panama	BBB-	Ba1	BB+	Uganda	B-	B1	B
Colombia	BB	Baa2	BB	Jordan	BB-	Ba3	BB-	Papua New Guinea	B-	B2	B+	Ukraine	CCC+	Ca	CCC-
Costa Rica	BB	Ba1	BB	Kazakhstan	BBB-	Baa2	BBB	Paraguay	BBB-	Ba1	BB+	United Arab Emirates	AA	Aa2	AA-
Croatia	A	A3	A-	Kenya	B	B3	B-	Peru	BBB-	Baa1	BBB	United Kingdom	AA-	Aa3	AA-
Cuba	NR	Ca	NR	Kuwait	AA-	A1	AA-	Philippines	BBB+	Baa2	BBB	Uruguay	BBB+	Baa2	BBB
Cyprus	A-	A3	A-	Kyrgyzstan	B+	B3	B	Poland	A-	A2	A-	USA	AA+	Aa1	AA+
Czechia	AA-	Aa2	AA-	Laos	CCC+	Caa2	CCC+	Portugal	A+	A3	A	Uzbekistan	BB	Ba2	BB
Democratic Republic of the Congo	B-	B3	NR	Latvia	A-	A3	A-	Puerto Rico	NR	Ba1	NR	Venezuela	CCC-	WR	WD
Denmark	AAA	Aaa	AAA	Lebanon	SD	C	RD	Qatar	AA	Aa2	AA	Vietnam	BB	Ba2	BB-
Dominican Republic	BB	Ba1	BB-	Lesotho	NR	NR	B	Republic of the Congo	CCC+	Caa2	CCC+	Zambia	CCC+	Caa2	D-

Source: Multiple public sources

As you can see, the ratings agencies mostly agree on their assessments of default risk, and sovereign ratings are correlated with the risk drivers (politics, corruption, violence, legal system) that we outlined in the last section. I do believe that ratings agencies, notwithstanding the critiques of bias and mis-measurement leveled against them, do a reasonably good job in their ratings assessments, but they are often slow to act, when confronted with change.

The sovereign CDS market offers a market-based alternative for measuring sovereign default risk, with investors making assessments of how much they would demand to insure against sovereign default in the form of (annualized) spreads. In the graph below, I list 10-year sovereign CDS spreads as of July 1, 2026:



Country	Sov CDS	Net of Swiss	Country	Sov CDS	Net of Swiss	Country	Sov CDS	Net of Swiss	Country	Sov CDS	Net of Swiss
Abu Dhabi	0.65%	0.52%	El Salvador	3.40%	3.36%	Kuwait	0.90%	0.77%	Russia	NA	NA
Algeria	1.21%	1.08%	Estonia	0.77%	0.64%	Latvia	0.84%	0.71%	Rwanda	2.96%	2.83%
Angola	5.03%	4.90%	Ethiopia	33.19%	33.06%	Lebanon	NA	NA	Saudi Arabia	0.97%	0.84%
Argentina	5.22%	5.09%	Finland	0.24%	0.11%	Lithuania	0.83%	0.70%	Senegal	10.05%	9.92%
Australia	0.20%	0.07%	France	0.59%	0.46%	Malaysia	0.66%	0.55%	Serbia	1.99%	1.86%
Austria	0.28%	0.15%	Gabon	7.44%	7.31%	Mexico	1.59%	1.46%	Slovakia	0.64%	0.51%
Bahrain	2.99%	2.86%	Ghana	3.70%	3.57%	Mongolia		-0.13%	Slovenia	0.79%	0.66%
Belgium	0.40%	0.27%	Germany	0.19%	0.06%	Morocco	1.29%	1.16%	South Africa	2.20%	2.07%
Brazil	2.21%	2.08%	Greece	0.63%	0.50%	Namibia	3.32%	3.19%	Spain	0.36%	0.23%
Bulgaria	0.78%	0.65%	Guatemala	1.66%	1.53%	Netherlands	0.20%	0.07%	Sri Lanka	NA	NA
Cameroon	5.85%	5.72%	Hong Kong	0.38%	0.25%	New Zealand	0.26%	0.13%	Sweden	0.18%	0.05%
Canada	0.29%	0.16%	Hungary	1.20%	1.07%	Nicaragua	5.30%	5.17%	Switzerland	0.13%	0.00%
Chile	0.80%	0.67%	Iceland	0.59%	0.46%	Nigeria	3.39%	3.26%	Thailand	0.67%	0.54%
China	0.52%	0.39%	India	0.85%	0.72%	Norway	0.15%	0.02%	Tunisia	7.07%	6.94%
Colombia	2.51%	2.38%	Indonesia	1.43%	1.30%	Oman	1.22%	1.09%	Turkey	2.83%	2.70%
Costa Rica	1.83%	1.70%	Iraq	3.86%	3.73%	Pakistan	4.87%	4.74%	Ukraine	NA	NA
Croatia	0.91%	0.78%	Ireland	0.31%	0.18%	Panama	1.65%	1.52%	United Kingdom	0.38%	0.25%
Cyprus	0.62%	0.49%	Israel	0.99%	0.86%	Peru	1.13%	1.00%	United States	0.56%	0.43%
Czech Republic	0.53%	0.40%	Italy	0.61%	0.48%	Philippines	1.02%	0.89%	Uruguay	0.79%	0.66%
Denmark	0.19%	0.06%	Japan	0.48%	0.36%	Poland	0.94%	0.81%	Venezuela	NA	NA
Dubai	0.99%	0.86%	Kazakhstan	1.22%	1.09%	Portugal	0.37%	0.24%	Vietnam	1.39%	1.26%
Ecuador	4.82%	4.49%	Kenya	4.21%	4.08%	Qatar	0.58%	0.45%	Zambia	3.03%	2.90%
Egypt	3.55%	3.42%	Korea	0.42%	0.29%	Romania	2.25%	2.12%			

Source: Bloomberg

Note that sovereign CDS spreads are available for only 84 countries, and that there are swaths of the world (Central and North Africa, frontier markets) where they are not available.

Country Composite Risk

When you lend money to governments or buy government bonds, sovereign default risk is your key concern, and both sovereign ratings and CDS spreads try to measure that risk. When running a business in a country, you are exposed to a much wider range of risks, and measuring exposure to those risks may require different measures. One alternative is **country risk scores**, where services evaluate how countries measure up on different risk drivers, and come up with composite scores for these countries. In the table below, I report the country risk scores from two services - Political Risk Services (PRS) and the Economist (EIU), at the start of July 2026:

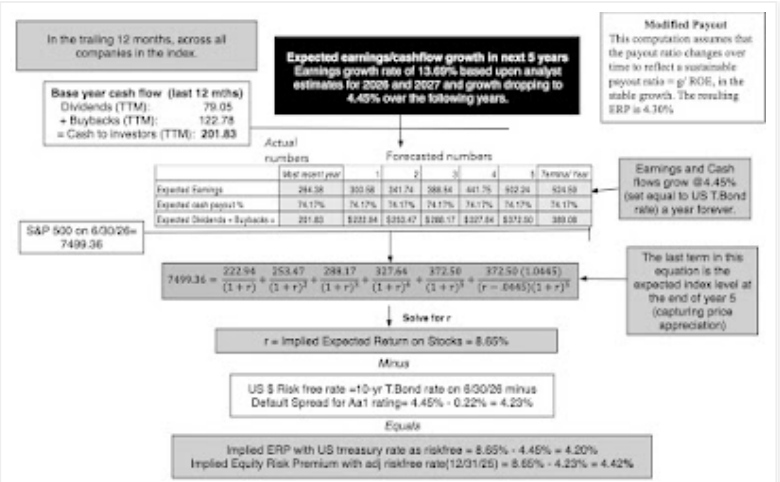


The Economist: Low (high) scores indicate less (more) country risk
PRS: High (low) scores indicate less (more) country risk

Sources: EIU (Economist) and PRS

The table illustrates three problems that you face with political risk scores. The first is that the scoring is idiosyncratic, with the Economist going from low scores for the safest countries to high scores for the riskiest, and PRS doing the reverse. The second is that each service picks different factors to consider, and different weightings, leading to scoring divergences that sometimes confound; PRS, for instance, ranks the United States as riskier than Ghana, on a composite risk basis. The third is that the scores, by themselves, are difficult to convert into inputs in financial analysis, either in cash flow or discount rate adjustment.

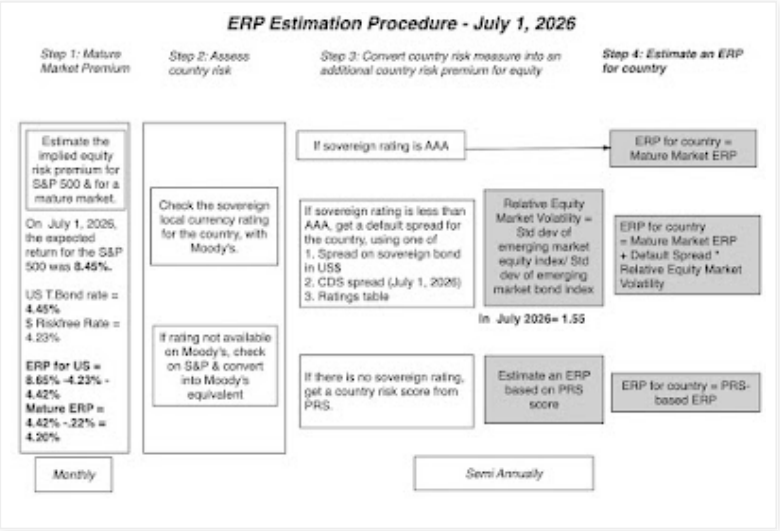
It is to combat the third problem that I started estimating country equity risk premiums, and while the details of the process and the data that I use have changed over the last three decades, the basic structure has remained unchanged. I start with an estimate of the **equity risk premium for a mature market**, and build a **country risk premium, if needed, for riskier countries**. Until 2025, I estimated the mature market premium by computing an implied equity risk premium for the S&P 500, and using that as the base, arguing that the US, as a Aaa rated country (at least according to Moody's), represented a mature market. The Moody's downgrade for the US, from Aaa to Aa1, has thrown a wrench into that approach, requiring adaptation. In response, I now start with an estimate of the implied ERP for the S&P 500, but then adjust that estimate for the default spread (based on the Aa1 rating) for the US, with the resulting values at the start of July 2026 below:



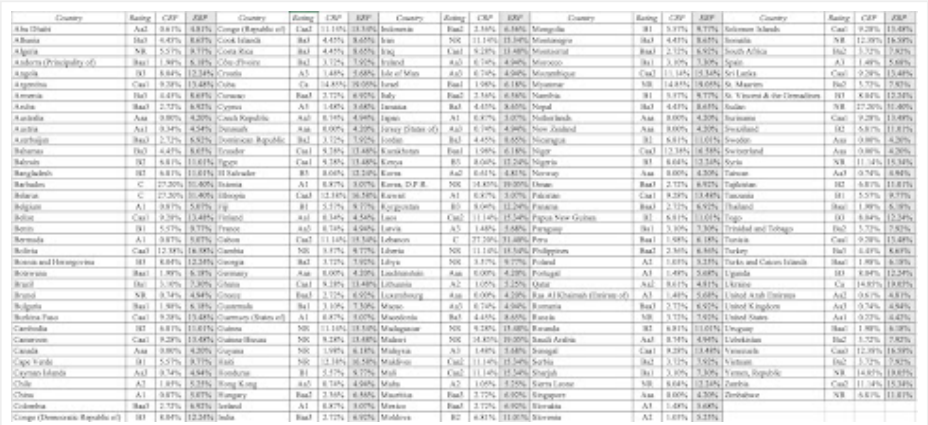
Spreadsheet: https://pages.stern.nyu.edu/~adamodar/pc/implprem/ERPJuly26.xlsx

As you can see, with the S&P 500 at 7499.36 on July 1, 2026, the implied equity risk premium for the United States is 4.42%, and netting out the default spread of 0.22% for the Aa1 rating yields a mature market premium of 4.20%.

To estimate country risk premiums, I *start with the sovereign ratings* for rated countries and convert those ratings into default spreads. To adjust for the higher risk associated with equities, relative to government bonds, I estimate a *composite measure of that relative risk*, by scaling the volatility in an emerging market equity index to the volatility in a emerging market government bond ETF, and scale the default risk up with this relative risk measure (1.55 in July 2026) to get country risk premiums:



For the two dozen countries that have no sovereign ratings, I adopt an even more makeshift approach, where I used political risk scores for these countries, and then looked for rated countries with similar scores. The table below has equity and country risk premiums, by country, for all of the countries that I evaluated in July 2026:

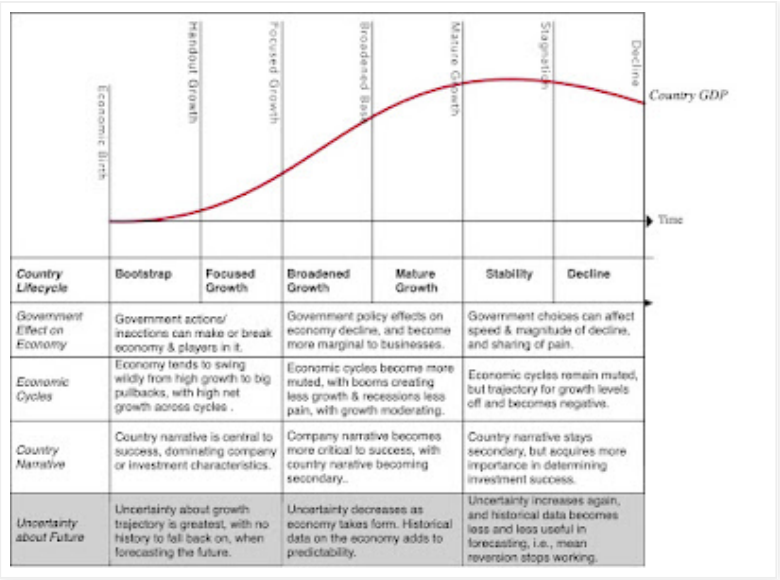


Download data

I did post an earlier version of this table a couple of weeks ago, but the numbers that I reported reflected in incomplete update of sovereign default spreads, and this table (and the data on my webpage) now reflect the corrected (and lower) spreads. *(As a solo act, I am deeply grateful for the checking that those who use my data do, and thankful when they point out mistakes that I have made.)*

Company Exposure to Country Risk

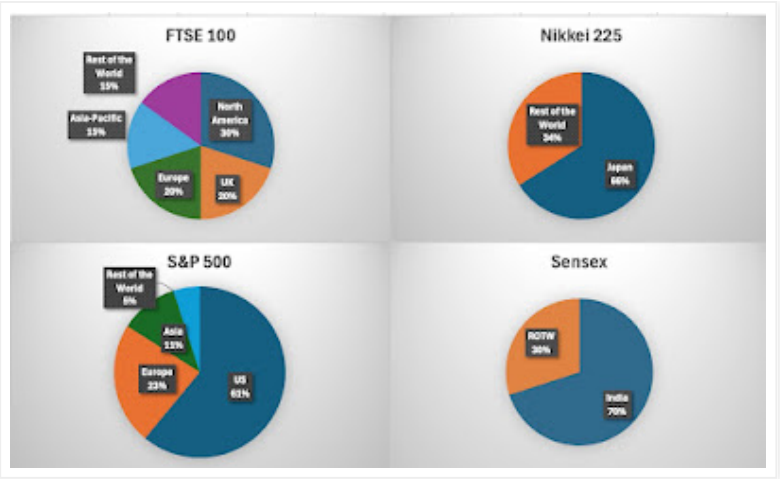
If you buy into my argument that every company has a narrative, and it is the narrative that drives its value, it is worth considering where country risk fits into that narrative. The answer, I believe, comes from looking at where the country in question falls in the life cycle:



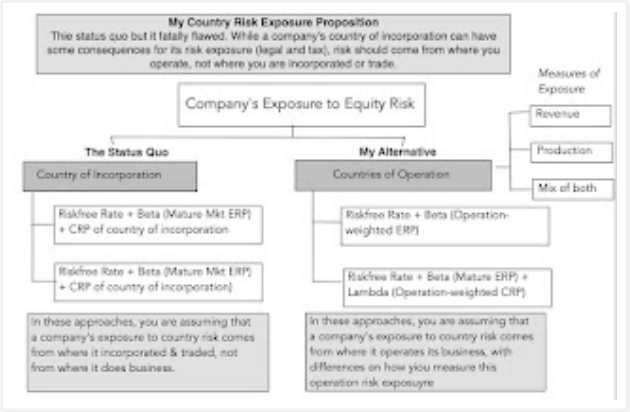
Country Lifecycle	Bootstrap	Focused Growth	Broadened Growth	Mature Growth	Stability	Decline
Government Effect on Economy	Government actions/ inactions can make or break economy & players in it.		Government policy effects on economy decline, and become more marginal to businesses.		Government choices can affect speed & magnitude of decline, and sharing of pain.	
Economic Cycles	Economy tends to swing wildly from high growth to big pullbacks, with high net growth across cycles .		Economic cycles become more muted, with booms creating less growth & recessions less pain, with growth moderating.		Economic cycles remain muted, but trajectory for growth levels off and becomes negative.	
Country Narrative	Country narrative is central to success, dominating company or investment characteristics.		Company narrative becomes more critical to success, with country narrative becoming secondary..		Country narrative stays secondary, but acquires more importance in determining investment success.	
Uncertainty about Future	Uncertainty about growth trajectory is greatest, with no history to fall back on, when forecasting the future.		Uncertainty decreases as economy takes form. Historical data on the economy adds to predictability.		Uncertainty increases again, and historical data becomes less and less useful in forecasting, i.e., mean reversion stops working.	

The message from this life cycle view is a sobering one, especially for those analyzing companies that operate in very risky countries, since the narratives for these companies implicitly or explicitly incorporate a country risk component. You cannot value a Venezuelan company without taking a strong view about Venezuela, or even an Indian and Brazilian company without an India or Brazil country story underpinning value. In contrast, you may be able to value US and European companies, without explicitly considering the evolution of country risk in those parts of the world.

When looking at an individual company, I believe that country risk exposure comes less from where the company is incorporated and more from where it operates. It is undeniable that companies around the world have substantial exposure outside their domestic markets, and that exposure has increased over time. In the graph below, I look at the revenue breakdown of companies in four indices - the S&P 500 (US), the FTSE 100 (UK), the Nikkei 225 (Japan) and the Sensex (India):



In every single index, companies that comprise that index get a significant portion of their revenues from outside the domestic market. Looking across sectors, exposure to foreign markets varies widely with technology companies often generating more than half of their revenues outside their domestic settings. I believe that equity risk premiums for companies should reflect exposure to foreign markets, though it is worth debating how best to weight that exposure - revenues work well for consumer product and service companies, production works better for natural resource companies and a mix of revenues and production may be the right choice for manufacturing companies:



With this framework, you can see why almost all analysts will confront country risk, sooner or later, no matter where they operate in the world and which companies they analyze.

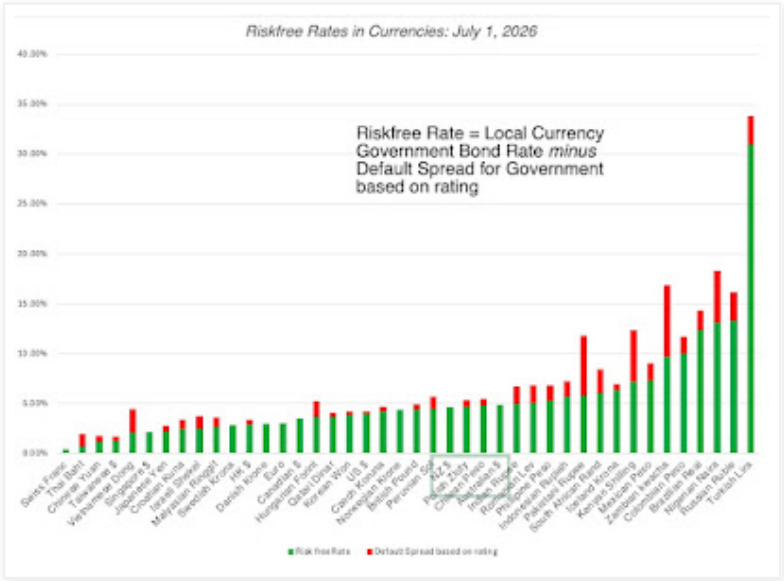
For companies, country risk will also come into play when faced with capital budgeting decisions, where they need estimates of hurdle rates for individual projects, to decide where to invest. For a multinational operating in many businesses, the project cost of equity will have to then also reflect the business the project is in, in addition to country risk. Thus, the cost of equity

for a Siemens Appliances for a project in India should reflect the beta for the appliance business, in addition to the country risk for India. In contrast, a Siemens power tool project in Hungary should be computed using the beta for an power tools project and the country risk for Hungary. It is also possible that country risk is not easy to isolate, if the production facilities are in one country but revenues are generated in another. If the Siemens appliance factory in India will be producing products that will be sold in Japan, should we be showing the country risk of India or Japan in the cost of equity calculation? The answer, as was the case in the earlier section on valuation, is that it depends on where the company sees risk coming from. If the risk is that production will be delayed or disrupted by political and economic risk in India, it is Indian country risk that should be looked at, but if the primary concern is that revenues in Japan will be volatile because of economic conditions there, it is Japanese country risk that matters more. If both risks are considerations, you should use a weighted average of Indian and Japanese country risk.

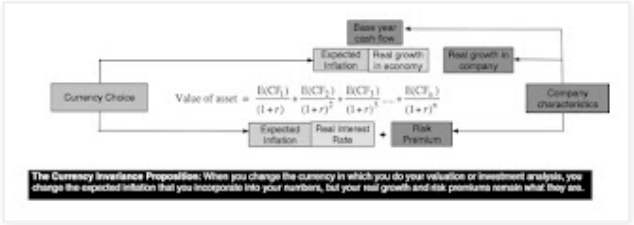
Currency Questions

For some of you, it may seem odd that I have spent almost an entire post talking about country risk without bringing up currencies. The reason is simple. Currencies are measurement mechanisms, and while they may be affected by the same political and economic factors that drive country risk, they don't determine country risk and in my view, should not command risk premiums, on their own.

It is true that hurdle rates are affected by both the equity risk premiums that you estimate and the riskfree rate that you use, and that riskfree rates vary across currencies. In the figure below, I estimate riskfree rates in about 40 currencies, where a local-currency government bond rate is present, and I adjust that government bond rate for the default risk of the government in question:

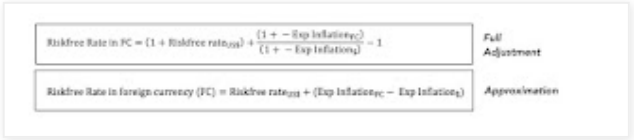


When estimating the cost of equity for a Turkish project or company in Turkish lira, we start with a riskfree rate in excess of 20% and build on it, by adding equity risk premiums to it, but the cost of equity for the same project or company in Euros will begin with a riskfree rate close to 3% (the German Euro bond rate) and arrive at a much lower number. While this may sound farfetched, the value that you derive for the project or company should be the same using either currency, if you are consistent about estimating your cash flows in the same currency:



Since much or almost all of the differences in riskfree rates come from inflation differentials, matching the high Turkish lira discount rate with a high growth in cashflows in Turkish lira, and the low Euro discount rate with the low growth in cashflows estimated in Euros will yield results that are consistent.

If you do want to estimate riskfree rates in currencies where there is either no local currency government bond that is traded or where you mistrust the government bond rate, because of light trading or government intervention, the fact that riskfree rate differences across currencies can be tied to differential inflation can be used for estimation; the riskfree rate in any currency can be computed from a base currency (dollar or Euro) riskfree rate and the difference in expected inflation between the local and base currencies:



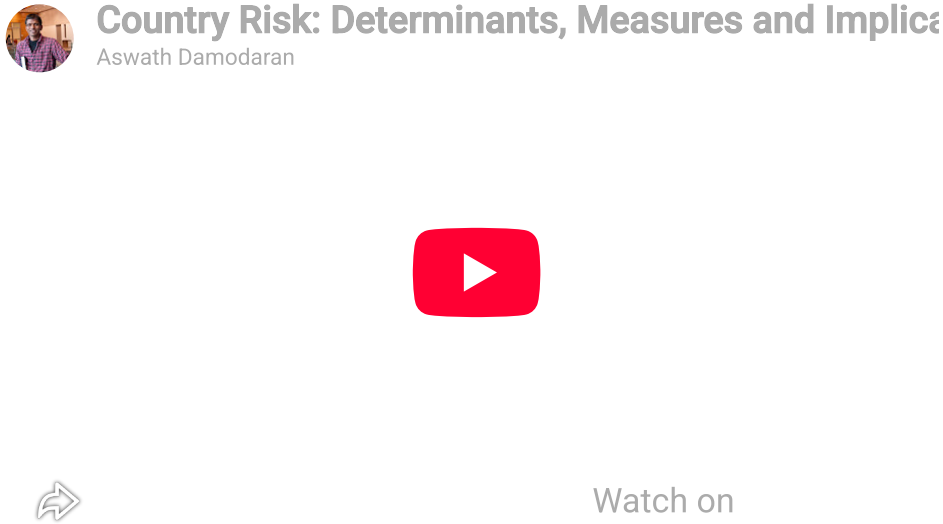
Put simply, if the expected inflation rate and riskfree rate in US dollars are 2.5% and 4% respectively, and the expected inflation rate in Brazil is 10.5%, the riskfree rate in Brazilian reais should be roughly 12%. The implication of this approach is that *currency pegs, when they do exist, will hold only if the inflation in the pegged currency matches or is close to the inflation in the index currency* to which it is pegged. It is true that the estimates of riskfree rates will only be as good as the expected inflation rates that are embedded in the estimation, but the good news is that being wrong on expected inflation will be largely offsetting, since both your cashflows and your discount rates will be wrong in the same direction; if you underestimate expected inflation, you will underestimate (overestimate) your riskfree and hurdle rates, but you will also underestimate (overestimate) your expected growth rate in cash flows.

Conclusion

One of the side effects of the rise of globalization is that there are fewer and fewer companies that are entirely local-country focused in both their revenues and production, and as a result, almost every business and investor is exposed to risk in other parts of the world. The problem with measuring country risk is that while its consequences are economic, it has its sources in history, politics and governance structures. The measures of country risk, whether they be entity-based like sovereign ratings, or market estimates like sovereign CDS spreads, reflect this interplay.

I confess that I have made simplistic assumptions and cut corners in my attempt to estimate equity risk premiums, by country, and there will be individual countries, perhaps even your own, where you might disagree with my assessments. As I noted earlier, my estimation approach remains a work-in-progress and I am always open to suggestions on how to estimate these premiums better, but keep in mind that whatever those improvements may be, they will have to work across 180 countries.

YouTube Video



Papers on country risk and equity risk premiums

1. Country Risk Premiums - The 2026 Edition (July 2026)
2. Equity Risk Premiums - The 2026 Edition (March 2026)

Data

1. Equity Risk Premiums, by country - July 2026

2. Implied Equity Risk Premiums for the S&P 500 (Annual since 1960 & Monthly since Sept 08)
3. Inflation-based riskfree rates, by currency - July 2026

Spreadsheet

1. Implied ERP for S&P 500 (July 2026)

Posted by Aswath Damodaran at 5:40 PM No comments:
Labels: Country Risk, Currencies, Equity Risk Premiums

Thursday, June 18, 2026

SpaceX, OpenAI and Anthropic: The S&P 500 Inclusion Question and Investment Consequences!

Over the last few weeks, attention has (rightly) been focused on three potentially trillion dollar companies all lined up to go public, and much of the discussion has been about what SpaceX, Anthropic and OpenAI are worth (and will be priced at). In parallel, there has been a debate about indices and index inclusion criteria, a usually bland topic, but one that has become heated on the questions of whether these new mega-cap additions to the market should be included in the S&P 500. While I remain open to arguments from both sides of this debate, much of it seems to come down on the side that the index should not include these companies, with different reasons offered.

I am skeptical, since I see a combination of hidden agendas and misguided views about investing behind each of the three groups that are most vehemently against inclusions. First, you have a cadre of active investors, many of whom have been left bruised by a losing battle that they have waged over the last two decades against passive investments (index funds and ETFs), who view inclusion in the index as a fait accompli, and present this as an added risk to passive investing that can be avoided by paying these professional money managers to avoid that risk. Second, you have investing experts and academics who claim to be looking out for for retail investors and retirees, and view including these big, money-losing companies in indices as dangerous for these small investors, partly because they may not be aware of their exposure and partly because they should not be investing in these types of companies. Third, you have politicians, normally not founts of investment wisdom, speaking out about how including these large companies in government pension funds will reward billionaires, who are the villains in their storylines. In this post, I will try to step back from the heat and try to cast some light on the question of index inclusion, starting with an understanding of how indices are constructed before

moving on to the roles they perform in markets and ending with a discussion of whether and how inclusion of these companies will affect the passive versus active investing debate.

Index Construction - Inclusion, Weights and Returns

Indices have been around almost as long as assets have been bought and sold in markets, but it is undeniable that the extraordinary growth of financial markets in the last few decades, across geographies and asset classes, has added rocket fuel both to the number of indices in existence as well as their visibility. But what is it that sets one index apart from another, and why do indices that purport to measure the same market sometimes move in different ways? To understand the answer to these questions, we need to deconstruct indices and see how they are put together:

- Constituents: The first and perhaps most critical determinant of an index are its constituents, and what determines their inclusion. Take, for instance, the S&P 500, which Standard & Poor's (its creator) describes as the "gauge of large-cap US equities", and is without doubt the most widely tracked and followed index in global markets. As its name indicates, this index has five hundred of the largest market-cap companies listed and traded in the United States, with caveats on inclusion relating to listing age (listed at least a year), liquidity (measured by looking at shares that are available for investors to trade in the market, i.e., free float) and profitability (positive profits in the four quarters leading into the listing). There are local indices that exchanges (NYSE, NASDAQ), equity markets in other geographies (the Bovespa for Brazilian stocks and the Sensex for Indian stocks) and individual sectors or industries. Across asset classes, there are indices for fixed income, as well as for real estate, fine art and crypto currencies.
- Weights: You can have two indices that contain the same companies that register very different results over time, depending on how these companies are weighted, with three common choices. The first is to weight every company in an index equally, with the benefit being simplicity, but the cost being that to the extent that companies in an index have very different sizes, the performance on an equally weighted index will not capture aggregate market performance, because it will be skewed towards smaller companies. The second, and one used by some older indices like the Dow 30, is price-weighting, where the companies with the highest price per share are weighted more than companies that have lower priced shares. I cannot think of a single redeeming quality to price weighting, since it measures very little of consequence, and suffers from breakdowns, right after stock splits. The third and most widely used mechanism for construction indices is market capitalization, with tweaks sometimes added on for float (traded shares). The S&P 500, as I noted earlier, uses market capitalization, based on free float, to weight companies and as a consequence, Meta punches in below its true weight, since the bulk of class B shares (which are voting shares held by Zuckerberg) are not counted, as does Walmart, where some family-controlled holdings are treated as non-traded.

- Index level mechanics: Once constructed, an index has to be measured, and to the extent that these indices are designed to capture market prices, the first step is creating a mechanism for converting market prices on the constituents to an index level. Consider, for instance, the S&P 500 which ended trading on June 15, 2026, at 7554.29, and relating that number to the market capitalization of the companies that make up the index. At close of trading on June 15, 2026, the cumulative float-adjusted market capitalization of the 500 companies in the index was $63,498.44 billion and the index units for the conversion can be computed as follows:

    Index units = Index level / Float-adjusted market capitalization = 7554.29/ 63498.44 = 0.1190

    Note that there is no intuitive significance to the index units standing alone, but its movements over time can be an indicator of changes happening at companies, because of issuances and stock buybacks, as well as changes in index constituents. If asked to compute earnings or dividends on the S&P 500, these index units come into play again, when converting the aggregated dividends and earnings across all of the S&P 500 companies into index dividends and earnings. In 2025, for instance, the aggregated dollar dividends on the S&P 500 was $664.90 billion, and multiplying that value by the index units (0.1190) yields an index dividend of 79.12 for the year.

- Price updating: While index levels are starting points, most investors track indices for changes in the index, with increasing stock prices translating into higher index values. Indices that track publicly traded stocks, like the S&P 500 and the Dow 30, should adjust instantaneously as the prices of their constituent companies change during the course of a trading day, making the index a real-time measure of market movements. Indices that capture only price changes miss the other component of returns on a stock, which is dividends, and constructing an index that incorporates dividends paid on a continuous basis does take work and requires assumptions about whether the dividends are reinvested in the index or extracted by investors. Though not as widely disseminated as the pure-price version, there is a variant of the S&P 500 that computes the total return on the index, with dividends included. Indices of assets that are not continuously traded, most notably real estate (like the S&P Case-Shiller home price index), try to overcome the absence of price data on the assets by extrapolating from the pricing of the subset of assets that get traded, leading to noisier estimates for index value and lags in price adjustment.
- Index changes (inclusions and exclusions): Even the best constructed indices have to confront change and have mechanisms to deal with that change that are transparent and quick to put into practice. Some of that change will come from companies being removed from public markets, either because they are acquired, taken private or because of bankruptcy. Some change will be caused by new companies being listed on the market or some will be created by changes in market cap in companies that bring them into contention for inclusion in the index, either because the market cap

has risen (making them large enough to qualify for a large cap index) or has dropped, removing them from large cap status. Since you do not want abrupt changes in the index level coming just from replacing a company with a low market cap with one with a much higher market cap, the adjustment has to come from changing the index units. Thus, assume that a company goes public with a trillion dollar market cap and that it will be replacing a company with a one-billion market cap, the adjusted index units for the S&P 500 will be as follows:

Index units = Index level / Float-adjusted market capitalization + Market cap of added firm - Market cap of eliminated firm = 7554.29/ (63498.44+ 1000 -1) = 0.1171

This will then percolate through into the index earnings and dividends estimates, for the index. Note that while the index level will be unchanged by the addition of the trillion dollar company, the other components that it brings with it, including higher growth and perhaps negative earnings, will alter the fundamentals of the index going forward.

With these index mechanics in mind, it is quite clear that if S&P does include SpaceX, OpenAi and Anthropic in the S&P 500 index, the index will not change at the time of the replacement, but it will change the index fundamentally going forward, bringing in more risk, a near term hit to earnings and perhaps a long term increase in growth.

The Index End Game

When indices were first created for markets, their primary purpose was to create composite measures of market performance, with a single number (the index value) capturing the performance of a much larger group of assets. Over time, though, the use of indices has expanded, first as proxies to assess the performance of active investors, to see whether they over or underperformed, and then as investment vehicles, with the advent and growth of index funds and ETFs.

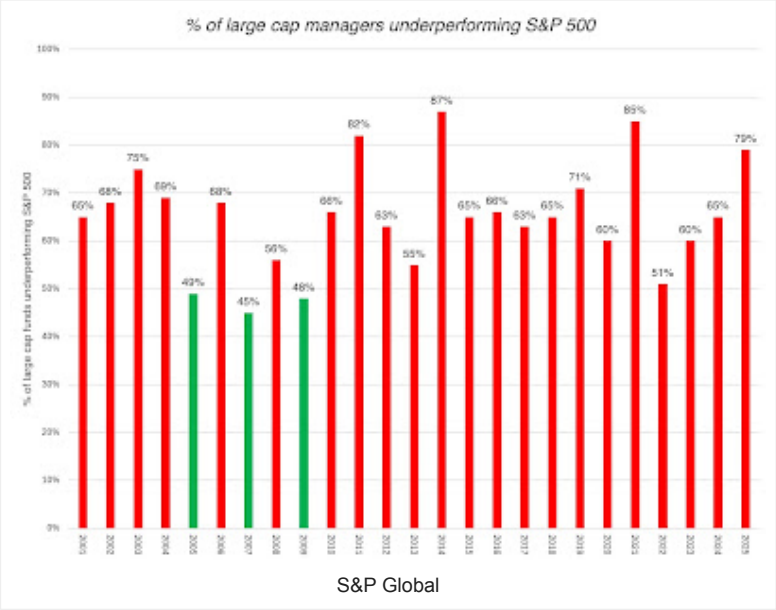
Measurement

I started in equity markets in 1981, and at that time, indices were primarily measures of market performance. At the time, for most investors without intraday access to markets and without financial news channels, news of market performance, on most days, was a snippet on the evening news, where the anchor would mention the market's change during the day, usually using the Dow 30 as a stand in for the market. Indices continue to perform the measurement role, though our access to data has changed dramatically, with real time updates on our devices occurring all through the day. From the measurement perspective, it is worth looking back at index construction and looking for indices that best capture what you are trying to measure. The reason that the S&P 500 has acquired primacy is that while it includes only 500 companies in a US equity market that has almost 6000 publicly traded companies, the fact that these are the companies with the largest market capitalization means that the index represents more than 80%

of the market capitalization of all US equities, and as a result, it is the single best proxy for aggregate equity market performance in the index. That said, a different framing of the measurement question can lead you to a different index choice. Thus, if you are trying to measure how the average US equity did during a period, you may be better served using an equal-weighted equity index for that measurement.

Performance Evaluation

Investors who trust professionals to manage their money, and pay them for their services, either as up-front entry fees or in annual management expenses, are entitled to wonder whether they are receiving a compensatory benefit, in the form of higher returns. It should not be surprising that comparing a mutual fund's returns to the returns on an index becomes a proxy for fund performance, and in the early years of performance evaluation, the S&P 500 became the default comparison index. Used in that context, one of the most jarring numbers in active investing is the percentage of active large cap funds that earn returns that are lower than the S&P 500, each year for the last two decades:



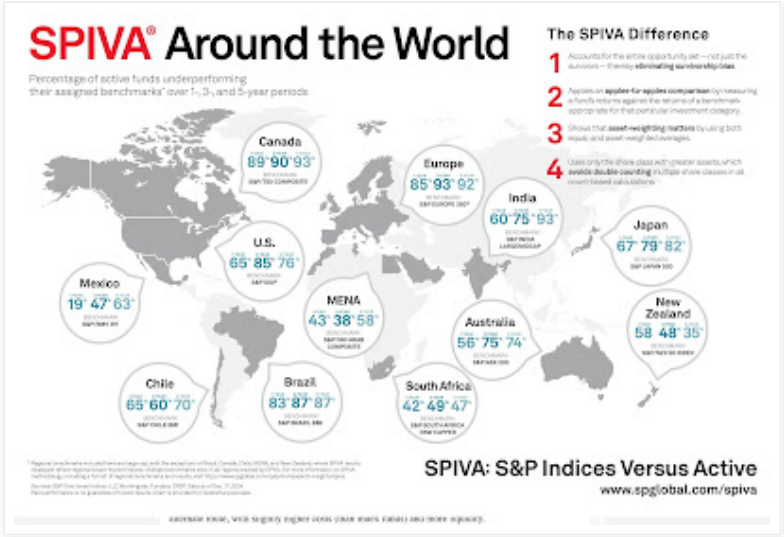
S&P Global

Over the last twenty five years, there have been three years where more than 50% of active funds have beaten the S&P 500 index, and barely so, and the extent of underperformance in the remaining years i staggering. The pushback from some active fund managers is that, given their investment styles, the S&P 500 is not the right index to use to judge them. Value fund managers, who invests in low-risk and high-dividend paying stock will argue that the portfolios they create are less risky than the S&P 500, making their lower returns more of a risk effect than underperformance. Academic studies that used risk and return models to tweak the index returns were quickly dismissed as being wrong, because the models that were used (the CAPM, the

APM, Multi-factor models) were flawed. This deadlock was broken by S&P, when it created SPIVA, where the returns earned by fund managers in any class (small cap vs large cap, growth vs value, domestic vs foreign) are compared to returns that investors could have earned by investing in index funds in the same class, and by Morningstar, using a variant of the same approach. Thus, a fund manager who invested in small-cap, high dividend paying stocks would see his or her returns compares to the returns you would have earned on a small-cap, value index fund, making it much more difficult to explain away underperformance. There are many reasons for the slippage in active investing's share of overall investing in the last two decades, but the SPIVA results are devastating and damning for any claim of active investing superiority. Here, for instance, are the results, by investment class, on the percentage of active fund managers who underperfomed index funds in their investment style, over the last decade, at the end of 2025:

	1-year	3-year	5 -year	10-year	15-year	20-year
All Domestic Funds	79.83%	80.40%	91.47%	90.43%	93.15%	95.01%
All Large-cap Funds	78.78%	66.84%	88.96%	85.59%	89.93%	92.89%
All Mid-cap Fnds	55.41%	62.93%	72.32%	81.14%	84.49%	89.74%
All Small cap Funds	40.65%	42.18%	62.67%	75.95%	89.90%	90.28%
All Mult cap Funds	68.03%	80.20%	90.03%	89.13%	92.25%	93.64%
Large-Cap Growth Funds	95.51%	56.33%	95.26%	91.67%	97.82%	99.55%
Large-Cap Core Funds	82.51%	83.12%	85.33%	93.40%	96.52%	94.81%
Large-Cap Value Funds	41.30%	80.51%	83.03%	88.45%	93.29%	87.01%
Mid-Cap Growth Funds	65.12%	58.14%	86.23%	65.79%	75.66%	86.76%
Mid-Cap Core Funds	46.02%	72.48%	59.79%	92.74%	90.00%	94.35%
Mid-Cap Value Funds	50.00%	67.86%	64.81%	84.48%	92.86%	90.11%
Small-Cap Growth Funds	34.08%	38.46%	79.70%	65.84%	87.91%	90.09%
Small-Cap Core Funds	42.57%	49.03%	50.00%	80.47%	91.58%	93.07%
Small-Cap Value Funds	44.90%	38.98%	47.30%	80.00%	95.28%	93.14%
Multi-cap Growth Funds	85.04%	70.13%	93.55%	93.64%	93.75%	94.61%
Multi-cap Core Funds	72.46%	84.00%	89.77%	95.88%	96.05%	94.92%
Multi-cap Value Funds	30.18%	72.35%	74.59%	88.33%	94.07%	91.76%
Real Estate Funds Funds	83.58%	88.61%	97.44%	84.04%	90.10%	91.14%

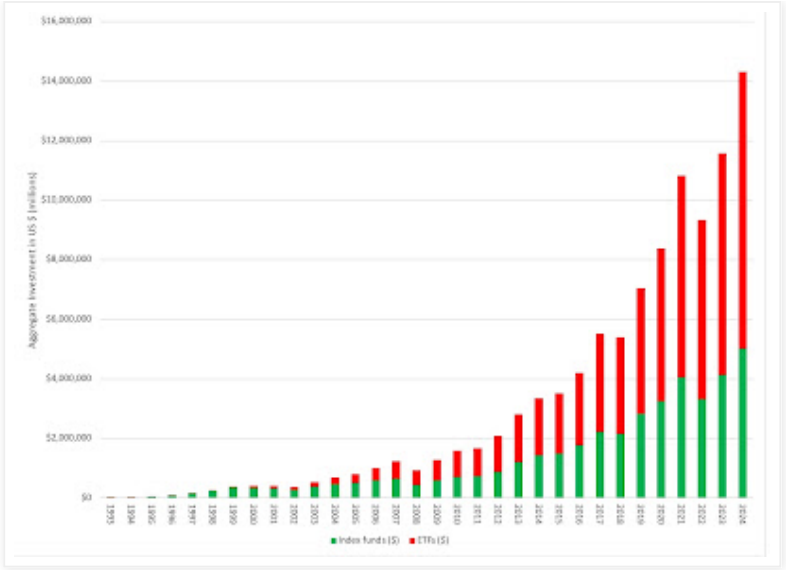
There is very little hope in these numbers, as fund managers underperform their respective indices in every single category, and by more, over longer periods. In fact, there is not a single fund group in any style that outperforms its respective index past ten years. For those of you who are reading this other geographies, and believing that it is different in your local markets, either because insiders have privileged access to information or market inefficiencies, SPIVA also tracks fund manager performance outside the United States, and reports similar results:



At this point, there is almost no counter to the argument that active investing collectively creates a drag on portfolio performance, and active investors seeking to defend the profession are left looking through the data entrails, hoping for niches where "alphas" exist. Thus, two decades ago, the notion that private equity investors and hedge funds were smarter than the market and could beat the market fueled a push of retirement and endowment money into these vehicles, but as they have become larger, they have come to resemble mutual funds, in terms of performance.

Investing Vehicles

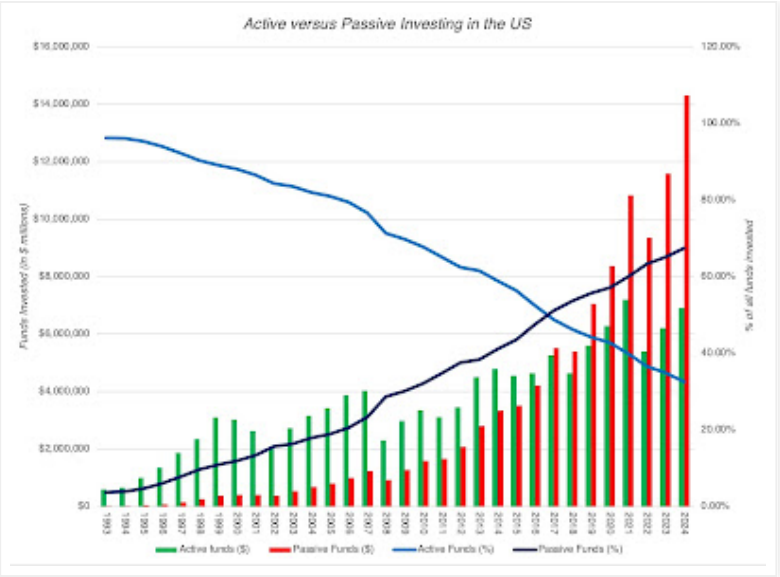
There is a third use of indices that, in my view, has overwhelmed the measurement and performance evaluation roles that they play, and it is that they have become vehicles for investing in the form of index funds and exchange-traded funds (ETFs). That possibility was already existent in 1981, but at that time, the only index with an index fund available to most investors was the S&P 500. Today, you can not only invest in index funds across geographies, sectors or sub-groups based on fundamentals (including volatility, size and earnings), but the exchange traded fund explosion has given you an alternate route, with slightly higher costs (than index funds) and more liquidity.



Vanguard, a pioneer when it listed the S&P 500 index fund in 1976, now lists more than a hundred ETFs and more than two hundred index funds, allowing investors to not only invest in almost any market, but also in the subsets (sectors, small companies etc.) that they chose to. The size of the index fund business and the fees its creates for the index creators has created some tensions in the process, and faced with a choice between a poorly constructed index that index fund investors would love to trade on and a better constructed one that investors find less attractive, it is possible and perhaps even likely that the fund creators will pick the former.

Indexing and Passive Investing - Unintended Consequences

There are two forces that the last section highlight that have played out in altering financial markets structurally and dramatically in the last two decades. The first is that *the underperformance of active investing has become easier to document and more visible for everyone to see*. The second is that investors who see this underperformance *have more passive investing vehicles in the form of index funds and ETFs accessible to them, and can move their money into them*. As a consequence, in the battle between active and passive investing for investor dollars, the fight is getting so one-sided that, if you were a referee, you would invoke the mercy rule and try to stop it:



It is undeniable that the rise of passive investing vehicles has empowered investors, who have gained in terms of choice and costs, but it has also created an existential crisis for active investors, and especially so for those who make a living out of managing other people's money. While some active investors hold on to hope - that they are better than the rest, that this is a cycle that will reverse, that some new technology (big data, AI) will save them- there are others who have taken a different tack. Mostly conceding that index funds and ETFs have outperformed fund managers, they have taken to arguing that the rise of passive investing is creating costs and effects that outweigh its benefits, and that action is needed urgently, though it is unclear from whom. In this section, I will highlight three of those effects.

1. The Index Inclusion Boost

The addition of a company to a well known or widely followed index (the S&P 500, Dow 30, MSCI Global) yields pluses, increasing its visibility to investors and potentially making it more investable, as well as increasing trading volume on the stock and making it more liquid. The question of how those benefits get priced in when a stock gets added to an index (and the costs of being removed from an index) have been studied over time, with a focus on additions to (and removals from) the S&P 500. As passive investing has grown, with more choices in index funds, it remains true that a significant percentage of passive investors hold S&P 500 index funds, and that, in the eyes of some, this should make inclusion in the S&P 500 index an even greater positive today than in decades past.

Looking across these studies, there seems to be a consensus that there is a bump up in the stock price from a company being added to the S&P 500, and bump down, when a company is deleted from the index, but disagreements both about the magnitude of the bump and whether it

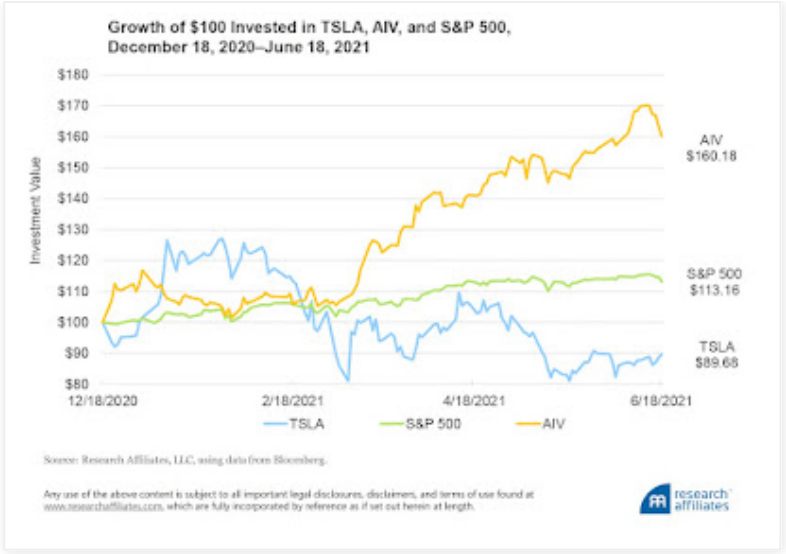
is permanent and transitory. The general sense that you get from studies is that *bump in stock prices from being included in the index has become smaller and more transitory over time, and in the last decade or two decades, it has largely disappeared.* To back this up, I look at one of the most complete studies that I have seen of the index inclusion question, where S&P took a look at the 715 companies added and 711 company deletions made to the S&P 500 between January 1995 and June 2021, and examine the excess returns in the days around the change:

	Additions to S&P 500			Removals S&P 500		
Time period	1995-1999	2000-2010	2011-2021	1995-1999	2000-2010	2011-2021
21 days before announcement date	0.61%	1.92%	2.30%	-0.99%	-6.69%	-1.76%
Announcement date to Effective date	8.32%	3.64%	-0.04%	-9.58%	-6.99%	0.06%
21 days after effective date	-4.49%	-1.69%	-1.69%	6.31%	4.59%	-2.15%
Total effect	4.09%	3.84%	0.53%	-4.83%	-9.23%	-3.81%

Link to study

As you can see, the positiv eprice effects of being added to the S&P 500 index have depleted over time, as have the negative price effects of being removed from the index. Note that this finding cuts against the argument that as passive investing has increased in the last two decades, the allure of being in the S&P 500 should also have gone up. Instead, as the value of funds indexed to the S&P 500 has surged over the last two decades, the effect of being added to or taken out of the index has become smaller, not larger, and there is evidence accumulating that companies that get added to the S&P 500 are more likely to underperform than outperform in the twelve months after the addition.

As the debate about whether SpaceX, OpenAI and Anthropic should be included in the S&P 500 index heats up, I would suggest that the evidence of a small and dissipating price effect of inclusion has to become part of the discussion. I am sure that there will be some who will disagree with me, but I don't think the price trajectories of any of these firms will be altered by whether they are included in or excluded from the S&P 500. For those who disagree with me, and believe that being added to index is bullish for investors in these companies, I would recommend that you look at the chart from this study which took a look at Tesla's stock price behavior before, during and after its replacement of Apartment Investment and Management (AIV) in the S&P 500 on December 18, 2020.



Tesla not only under performed the S&P 500 in the months after its inclusion in the index, but massively underperformed the company (AIV) that it replaced in the index.

2. Momentum versus Fundamentals

As the funds invested in index funds and ETFs has surged, the argument that some are making is that being added to an index gives you a boost, largely because of index fund flows to companies in that index, and more of a boost if you are a large cap company. Since the money flows from other stocks, this line of thought also suggests that fundamentals will receive less attention and disconnect more from prices, especially because there are fewer active investors doing research and looking for market inefficiencies. In addition, they note that since indices are mostly market cap weighted, this momentum benefits larger market cap companies, in effect allowing them to become larger. As evidence in favor of this argument, they point to the fact that markets have become top heavy, where a few winners are carrying the entire market, that the small cap premium, an enduring feature of equity markets in the twentieth century, has largely disappeared in this one and the dominance of momentum in investing success in the last decade. I concede that these phenomena are consistent with the "passive investing feeds momentum" story, but I am skeptical that passive investing is the cause for the following reasons:

1. Momentum can cut in both directions: It is true that funds flowing into index funds flow into the companies in that index, with more flowing into large cap companies, but it is also true that funds can flow out of index funds, and when that happens, the momentum can cut in the other direction. In fact, the conclusion is that inclusion in a widely-tracked index (like the S&P 500) will increase intraday and short term volatility, but not price levels. In fact, the fading price bump from being added to the S&P 500 that we noted in the last section is an indication that the market does not buy into the momentum story.

2. Winner-take-all economics: If momentum is the reason for the big companies winning, there should be divergence between small and big companies on how fundamentals get priced. Put simply, you should see the pricing metrics (PE ratios, EV to EBITDA) for large cap companies increasing relative to small cap companies, as passive investing surges. Looking back at the fading small cap effect and top-heavy markets of the last decade or two, I would note that not all large cap companies have been winners, and the winning large cap companies have delivered a disproportionate portion of increased earnings. In the context of the Mag Seven, I have talked about how technology and disruption has changed more industries into winner-take-all businesses, with a few companies dominating these businesses, and why that phenomenon will play out in markets as well.
3. Active investing and equity research: In my view, the notion that most analysts and active investors are looking for market inefficiencies and seeking out information strikes me as misplaced. Much of active investing is built around publicly available information and a belief in the power of mean reversion, not original research and seeking information. It is true that there is a subset of active investors and equity research analysts who contribute to making prices more informative, but that subset is a small one, and one that is better equipped to survive the passive investing shift.

Doomsday stories about how passive investing is making markets less efficient and less inclined to reflect fundamentals strike me as overwrought, and while active investing will continue to lose market share, and deservedly so, it will not disappear. Since these stories are often being told by fund managers who not so long ago spoke contemptuously about efficient markets as an academic fever dream, they also strike me as both hypocritical and self-serving.

Even if you accept the argument that passive investing is making markets less efficient and more momentum-driven is true, I am unsure about the implications for investing. Asking individual investors, retirees and endowment funds to pay fees to professionals to manage their money while underperforming indices, in service to the larger cause of market efficiency is tone deaf and a non-starter. In fact, any endowment or pension fund manager who uses this argument to steer endowment funds to active money managers would be in violation of his or her fiduciary responsibility. Perhaps, the argument is being made to regulators to restrict index funds (on which indices they can index, how much money they can manage), I can see why active money managers may be in favor, because I understand that they are trying to protect their livelihood, but they should dispense with any talk about protecting individual investors or making markets more efficient.

3. Hidden risks

An undercurrent in some of the opinion pieces that I have read about why SpaceX should not be included in the S&P 500, written by investment experts and academics, is that it will expose retail investors and retirees to risks that they are unaware they are taking, or even if made aware of the fact, should not be taking in the first place. In particular, these opinion-writers seem to be arguing that the risks associated with investing in a big, money-losing companies (like SpaceX,

and presumably OpenAI and Anthropic, when they go public) are so large that individual investors and retirees would not invest in these companies, and even they would, they should not be allowed to do so. I find this chain of reasoning to be both misguided and condescending, and reflective of misconceptions that are deeply and widely held in the investment expert class:

1. Risk and Diversification: Is it true that individual investors, if made aware of the companies that they owned in index funds, would blanch at the risks that they have exposed to in individual holdings? Investing just in a portfolio of a few companies like SpaceX would be imprudent, but an investor in a S&P 500 index fund is far less exposed to underperforming the market than the typical active money manager who either over invests in SpaceX (if it goes down) or chooses not to invest in it (if it goes up).
2. Smart and Stupid Money: While the investment experts and academics who push to protect retail investors and retirees from their own mistakes will never put into words this belief, implicit in this push is the view that these small investors are uninformed and naive, and will be exploited by smart money (institutional investors and hedge funds). The notion that the smart money will know whether SpaceX (and companies like it) is overvalued or under valued, and is positioned to time investments better is fanciful, since institutional investors are more traders than investors, making them market followers, not leader.
3. Good businesses and good investments: The weakest link in the argument against putting your money in money-losing companies is the implicit belief that money-making companies are good (safe) investments and that money-losing companies are bad (risky) ones. I will wager than an investor who was constrained to invest only in money-making businesses in the last two decades would have under performed an investor operating without those constraints, even after adjusting for risk. At the right price, a money-losing company can be a good investment and at the wrong price, a company with solid and stable profits can be a bad investment.

I am generally skeptical of attempts to protect individual or retail investors from their own mistakes and decisions, since more damage has been done to this group by those claiming try to help and protect them over time than by those who are out to exploit them.

Conclusion

In the week prior to the SpaceX IPO, S&P removed some of the suspense in the question of whether the company would be included in the index by announcing that they would stick with their requirement that a company be listed and traded at least a year before becoming eligible for index inclusion. That decision also means that OpenAI and Anthropic, if they do go public this year, will also have to wait a year for consideration. I am glad that S&P is not changing the rules to allow these companies to jump the queue to get into the index, but I hope that it is not framed as a decision that was taken to protect investors or in the hope that these companies would become magically money making, better governed and with working business models. The truth is that a year after they list and start trading, these three companies will still be money losing businesses, with business models that are still works in progress and will remain corporate

governance horror stories. S&P needs the time to manage the transition of three trillion-dollar companies into the index, even as it confronts the challenge of claiming to be a large cap index that does not include three of the largest market cap stocks in the market. As for the companies (SpaceX, OpenAI and Anthropic), I will wager that they will lose little in market momentum from not being included in the index, and that their price paths will be determined by how the AI story continues to play out in terms of both substance (growth, unit economics, reinvestment) and perception (hype and momentum). The bottom line is that S&P needs these companies in its index more than they need to be in the index, with the consequence that the companies will not go out of their way to meet index requirements that they feel are costly to them, and that if there is any bending, it will be S&P that does it.

YouTube Video



The Indexing Question: Should SpaceX, OpenAI an

Aswath Damodaran





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